

A TEMPERED DROP LEMMA FOR BAYESIAN VARIABLE SELECTION

BY YVANN LE FAY^{1,a}

¹CREST, ENSAE, IP Paris, ^ayvann.lefay@ensae.fr

Sequential Monte Carlo samplers for the Bayesian variable selection model of Yang, Wainwright and Jordan (2016) travel along the tempering path obtained by raising the marginal likelihood to a power $\lambda \in [0, 1]$, and deliver, besides the posterior, the normalising constant that a Markov chain alone does not give. Their cost is governed by the mixing of the Metropolis–Hastings kernel at *every* intermediate temperature, whereas Yang, Wainwright and Jordan (2016) bound the spectral gap at $\lambda = 1$ only, by a canonical path ensemble directed at the true model γ^* . We ask what survives at $\lambda < 1$, where the sparsity prior is untempered while the likelihood is damped, so that the mode of the target slides from the null model at $\lambda = 0$ to γ^* at $\lambda = 1$.

1. Introduction. Yang, Wainwright and Jordan (2016) — YWJ throughout — study high-dimensional Bayesian variable selection with a truncated sparsity prior, and prove two things about it. The posterior concentrates on the true model, and a particular Metropolis–Hastings chain targeting that posterior mixes in a number of steps linear in the number p of covariates, up to logarithmic factors. The second statement is the delicate one, and they show by example that it does not follow from the first: a posterior can be sharply concentrated and still be, for a local chain, effectively multimodal. Their proof of it is a canonical path argument, and the paths are the trajectories of a greedy variable-selection algorithm, all of them directed at the true model γ^* .

This note asks what becomes of that argument when the posterior is tempered: when the marginal likelihood is raised to a power $\lambda \in [0, 1]$ and the sparsity prior is left alone, so that the target interpolates between the prior at $\lambda = 0$ and the posterior at $\lambda = 1$.

Why the tempered targets, and not only the posterior. Two quantities are wanted from this model, and only one of them is delivered by a Markov chain. The first is the posterior over models, which an MCMC run approximates directly. The second is the normalising constant

$$Z = \sum_{\gamma} \pi(\gamma) \mathcal{L}_n(Y | \gamma),$$

the sum running over the models the prior retains — the marginal likelihood of the data under the whole hierarchical model — which a single MCMC run does not give at all: the chain sees ratios of posterior masses, and Z cancels from every one of them.

Z is a function of the hyperparameters (g, κ) of (5), and the theory does not choose them: Assumption 3.3 constrains κ and the exponent $\alpha = \log(1 + g)/(2 \log p)$ only through $\kappa \geq 2$ and $\kappa + \alpha \geq 8(L + \tilde{L}) + 2$, and $L, \tilde{L}$ are unknown. They must therefore be chosen from the data, by maximising $Z(g, \kappa)$ or averaging against a hyperprior — a marginal-likelihood computation over a grid.

2. Setting. We keep the model of YWJ and their transition function, and change one thing only: the marginal likelihood is raised to the power λ . Every other quantity that occurs is written out rather than named.

2.1. *The model.* Let $Y \in \mathbb{R}^n$ be a response vector and $X \in \mathbb{R}^{n \times p}$ a design matrix whose columns $X_1, \dots, X_p$ are normalised so that $\|X_j\|_2^2 = n$. The data are generated by

$$(1) \quad Y = X\beta^* + w, \quad w \sim \mathcal{N}(0, \sigma_0^2 I_n),$$

with $\beta^* \in \mathbb{R}^p$ and $\sigma_0 > 0$ unknown.

A subset of covariates is encoded by a binary vector $\gamma \in \{0, 1\}^p$, and we adopt the dualistic view of YWJ, using the same letter for the vector and for the subset $\{j \in [p] : \gamma_j = 1\}$ of $[p] := \{1, \dots, p\}$ that it indicates, so that $j \in \gamma$, $\gamma \cup \{j\}$, $\gamma \subseteq \gamma'$ and $|\gamma| = \sum_j \gamma_j$ all have their obvious meaning. We write X_γ for the submatrix of X with columns indexed by γ and

$$(2) \quad \Phi_\gamma := X_\gamma(X_\gamma^\top X_\gamma)^{-1} X_\gamma^\top, \quad \Phi_\emptyset := 0,$$

for the orthogonal projection onto their span. We use repeatedly, and without further comment, the fact that if $\gamma \subseteq \gamma'$ then $\Phi_{\gamma'} \Phi_\gamma = \Phi_\gamma$, so that $\Phi_{\gamma'} - \Phi_\gamma$ is itself an orthogonal projection, and that $\|\Phi_\gamma v\|_2 \leq \|v\|_2$ for every v .

Fix a threshold $C_\beta > 0$. The set of *influential* covariates is

$$(3) \quad S := \{j \in [p] : |\beta_j^*| \geq C_\beta\}, \quad S^c := [p] \setminus S, \quad s^* := |S|,$$

For a non-empty $M \subseteq \gamma^*$ we write

$$(4) \quad \overline{\beta^2}(M) := \frac{1}{|M|} \sum_{j \in M} |\beta_j^*|^2 \geq C_\beta^2$$

for the average squared coefficient over M , and γ^* denotes the binary vector indicating S , so that $X_S \beta_S^* = X_{\gamma^*} \beta_{\gamma^*}^*$; we use the two notations interchangeably, as YWJ do. The coefficients $\beta_{S^c}^*$ off the support need not vanish; they are only required to be small, which is Assumption 3.1 below.

The Bayesian model puts a flat prior $\pi(\phi) \propto 1/\phi$ on the noise precision, Zellner's g -prior $\beta_\gamma | \gamma, \phi \sim \mathcal{N}(0, g\phi^{-1}(X_\gamma^\top X_\gamma)^{-1})$ on the regression coefficients, and the truncated sparsity prior

$$(5) \quad \pi(\gamma) \propto p^{-\kappa|\gamma|} \mathbb{I}[|\gamma| \leq s_0],$$

on the model space $\mathcal{M}(s_0) := \{\gamma \in \{0, 1\}^p : |\gamma| \leq s_0\}$, where $\kappa > 0$ and $s^* < s_0 < n$. Integrating out (β_γ, ϕ) gives the marginal likelihood of their equation (13),

$$(6) \quad \mathcal{L}_n(Y | \gamma) \propto \frac{1}{\|Y\|_2^n} \cdot \frac{(1+g)^{-|\gamma|/2}}{\{1+g(1-R_\gamma^2)\}^{n/2}}, \quad R_\gamma^2 := \frac{Y^\top \Phi_\gamma Y}{\|Y\|_2^2}.$$

We assume $Y \neq 0$, an event of probability one, so that R_γ^2 is well defined.

2.2. *The tempered posterior.* For $\lambda \geq 0$ define

$$(7) \quad \pi_n^{(\lambda)}(\gamma | Y) := \frac{\pi(\gamma) \mathcal{L}_n(Y | \gamma)^\lambda}{\sum_{\bar{\gamma} \in \mathcal{M}(s_0)} \pi(\bar{\gamma}) \mathcal{L}_n(Y | \bar{\gamma})^\lambda}, \quad \gamma \in \mathcal{M}(s_0).$$

Since $\mathcal{M}(s_0)$ is finite and $\mathcal{L}_n(Y | \cdot) > 0$ on it, the denominator lies in $(0, \infty)$ and $\pi_n^{(\lambda)}(\cdot | Y)$ is a well-defined probability measure with full support on $\mathcal{M}(s_0)$, for every real λ . At $\lambda = 0$ it is the sparsity prior π ; at $\lambda = 1$ it is the posterior $\pi_n(\cdot | Y)$ of YWJ.

For every $\lambda \geq 0$ and every $\gamma, \gamma' \in \mathcal{M}(s_0)$,

$$(8) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = (|\gamma'| - |\gamma|)(\kappa + \lambda\alpha) \log p + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2}.$$

The first term of (8) is a *penalty*, affine in λ with intercept $\kappa \log p$ because the sparsity prior is untempered while the Occam factor $(1 + g)^{-1/2}$ is; the second is a *fit* term, proportional to λ and switched off at $\lambda = 0$. Every statement below compares the two, and the whole difference with the untempered argument of YWJ is that at $\lambda < 1$ the fit term is damped while $\kappa \log p$ is not. For $|\gamma'| - |\gamma| = 1$, (8) reads

$$(9) \quad \log \frac{\pi_n^{(\lambda)}(\gamma \cup \{j\} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} = \lambda \log \frac{\mathcal{L}_n(Y | \gamma \cup \{j\})}{\mathcal{L}_n(Y | \gamma)} - \kappa \log p.$$

3. Known results. We restate the assumptions of YWJ, and add one, Assumption 3.6, that their Lemma 4 uses without stating. Two of them must be imposed slightly above the truncation level s_0 , because Lemma A.1 below compares a model $\gamma \in \mathcal{M}(s_0)$ with $\gamma \cup \gamma^*$, whose size may exceed s_0 . We therefore impose their Assumptions B and D at the level

$$(10) \quad s := s_0 + s^* \leq 2s_0,$$

and work with the model class $\mathcal{M}(s)$. Since $s^* < s_0$ this strengthens their hypotheses by a factor at most two in the model size, and by nothing else.

ASSUMPTION 3.1 (Size of β^* ; YWJ, Assumption A). There is a constant $\tilde{L} < \infty$ with

$$\|X\beta^*\|_2^2 \leq g\sigma_0^2 \log p, \quad \|X_{S^c}\beta_{S^c}^*\|_2^2 \leq \tilde{L}\sigma_0^2 \log p.$$

ASSUMPTION 3.2 (Design; YWJ, Assumption B at level s). There are constants $\nu \in (0, 1]$ and $L < \infty$ with $L\nu \geq 4$ such that, with $Z \sim \mathcal{N}(0, I_n)$,

$$\min_{|\gamma| \leq s} \lambda_{\min}(n^{-1}X_\gamma^\top X_\gamma) \geq \nu, \quad \text{and} \quad \mathbb{E}_Z \left[\max_{|\gamma| \leq s} \max_{k \in [p] \setminus \gamma} n^{-1/2} |\langle (I - \Phi_\gamma)X_k, Z \rangle| \right] \leq \frac{1}{2} \sqrt{L\nu \log p}.$$

ASSUMPTION 3.3 (Hyperparameters; YWJ, Assumption C). $g \geq n$, $\kappa \geq 2$, and, with α defined by

$$(11) \quad \alpha := \frac{\log(1+g)}{2\log p}, \quad \text{equivalently} \quad \frac{1}{2} \log(1+g) = \alpha \log p,$$

$$\alpha \geq \frac{1}{2} \text{ and } \kappa + \alpha \geq 8(L + \tilde{L}) + 2.$$

The normalisation $\|X_j\|_2^2 = n$ forces $\lambda_{\min}(n^{-1}X_\gamma^\top X_\gamma) \leq 1$, so $\nu \leq 1$ is not a restriction, and $L\nu \geq 4$ is what YWJ's proof of Lemma 3.1 needs to make the deviation exponent of $\mathcal{A}_n$ at least $\frac{1}{2}$; together they give $L \geq 4$, which is used only through Corollary 4.2. Assumption 3.3 is their $\kappa \geq 2$, $\alpha \geq \frac{1}{2}$, $g \asymp p^{2\alpha}$, $g \geq n$ and $\kappa + \alpha \geq 4(L + \tilde{L}) + 2$, with two changes. The constant 4 is 8 because our overfitted bound (27) carries $8(L + \tilde{L}) \log p$ where theirs carries $4(L + \tilde{L}) \log p$; and α is defined by (11) rather than declared to satisfy $g \asymp p^{2\alpha}$. The second change costs nothing and removes a genuine ambiguity: the underfitted case needs $\frac{1}{2} \log(1+g) \leq \alpha \log p$ and the overfitted case needs $\frac{1}{2} \log(1+g) \geq \alpha \log p$, and an $\asymp$ supplies neither with a constant of 1. With (11) both hold as equalities, and we write $\alpha \log p$ for $\frac{1}{2} \log(1+g)$ throughout: every occurrence of the Occam factor of the g -prior below is in that form, and every inequality in which α occurs is exact. The requirement $g \geq n$ is the one under which YWJ state their event $\mathcal{D}_n$ below; it holds automatically when $\alpha \geq \frac{1}{2}$, since $p \geq n$, and it is satisfied by each of the choices $g = n$ and $g = \max\{n, p^2\}$ that they discuss in their Section 3.1.

ASSUMPTION 3.4 (Sparsity control; YWJ, Assumption D at level s). With $\omega(X) := \max_{|\gamma| \leq s} \|(X_\gamma^\top X_\gamma)^{-1} X_\gamma^\top X_{S \setminus \gamma}\|_{\text{op}}^2$,

$$(12) \quad \max\{1, (4\nu^{-2}\omega(X) + 1)s^*\} \leq s_0 \leq s \leq \frac{1}{256} \left\{ \frac{n}{\log p} - 32\tilde{L} \right\}.$$

The only difference with YWJ here is the constant 4 in place of their 2 on the left, which buys the margin that Lemma A.4b needs and that their proof of the saturated case does not in fact have (see Remark A.1); on the right they have $\frac{1}{32}\{n/\log p - 8\tilde{L}\}$, at the level s_0 .

ASSUMPTION 3.5 ($\beta_{\min}$ condition; YWJ, equation (23)). $n\nu^2 C_\beta^2 \geq 128(L + \tilde{L} + \alpha + \kappa)\sigma_0^2 \log p$.

This is their $\beta_{\min}$ condition $C_\beta^2 \geq 128\nu^{-2}(L + \tilde{L} + \alpha + \kappa)\sigma_0^2 \log p/n$, verbatim, with the factor ν^2 folded in. It is worth separating the two uses made of it below, because they are unrelated. The part $128(L + \tilde{L})$ controls the noise carried by a rank-one increment, and is what Lemma A.2b consumes; the part $128(\alpha + \kappa)$ never meets the noise at all, and is what raises the fit term of Theorem 4.1 above the Occam factor of the g -prior — that is, it is one of the two hypotheses that make Corollary 4.1 non-vacuous, by Corollary 4.2. The other is the following, which YWJ do not state.

ASSUMPTION 3.6 (Sample size). $n\nu \geq 32s^*(L + \tilde{L} + \alpha + \kappa) \log p$.

LEMMA 3.1 (Good event; YWJ, Lemma 3). *Define*

$$(13) \quad \begin{aligned} \mathcal{A}_n &:= \left\{ \max_{\gamma_2 \subset \gamma_1 \in \mathcal{M}(s)} \frac{w^\top (\Phi_{\gamma_1} - \Phi_{\gamma_2})w}{|\gamma_1| - |\gamma_2|} \leq L\sigma_0^2 \log p \right\}, & \mathcal{B}_n &:= \left\{ \max_{\gamma \in \mathcal{M}(s)} \frac{w^\top \Phi_\gamma w}{|\gamma|} \leq 8\sigma_0^2 \log p \right\}, \\ \mathcal{C}_n &:= \left\{ \left| \frac{\|w\|_2^2}{n\sigma_0^2} - 1 \right| \leq \frac{1}{2} \right\}, & \mathcal{D}_n &:= \left\{ \frac{\|Y\|_2^2}{g} \leq (2\log p + 3)\sigma_0^2 \right\}. \end{aligned}$$

Under Assumptions 3.1 to 3.4, $\mathbb{P}(\mathcal{A}_n \cap \mathcal{B}_n \cap \mathcal{C}_n \cap \mathcal{D}_n) \geq 1 - 6p^{-c}$ for a universal constant $c > 0$.

This is Lemma 3 of YWJ with their model class $\mathcal{M}(s_0)$ replaced by $\mathcal{M}(s)$, which their proof allows because the truncation level enters it only through that class.

From here on, every statement made “on the good event” is a deterministic statement about a fixed realisation of (Y, w) lying in $\mathcal{A}_n \cap \mathcal{B}_n \cap \mathcal{C}_n \cap \mathcal{D}_n$. No further probabilistic argument occurs.

Under Assumptions 3.1 to 3.5, YWJ prove two separate statements about the posterior $\pi_n(\cdot | Y) = \pi_n^{(1)}(\cdot | Y)$: that it concentrates on γ^* , and that the Metropolis–Hastings chain targeting it mixes in $\tilde{\mathcal{O}}(ps_0^2)$ steps.

3.1. Posterior concentration.

THEOREM 3.1 (Posterior concentration; YWJ, Theorem 1). *Suppose Assumption 3.1 and Assumption 3.3; Assumption 3.2 at the level $s = 2(\kappa + \alpha + \tilde{L} + 1) \max\{1, s^*\}$; the*

sparsity prior (5) left untruncated, $s_0 = p$, together with

$$\max\{1, s^*\} \leq \frac{1}{32} \left\{ \frac{n}{\log p} - 8\tilde{L} \right\} \quad \text{in place of Assumption 3.4,}$$

$$\text{and} \quad C_\beta^2 \geq \frac{128}{\nu^2} (L + \tilde{L} + \alpha + \kappa) \frac{\sigma_0^2 \log p}{n}.$$

Then $\pi_n(\gamma^* | Y) \geq 1 - c_1 p^{-1}$ with probability at least $1 - c_2 p^{-c_3}$, the c_i being universal.

3.2. The Metropolis–Hastings chain. The sampler is that of YWJ, and it depends on the distribution it targets only through the acceptance ratio. We therefore set it up once, for an arbitrary probability measure μ with full support on $\mathcal{M}(s_0)$, and carry no temperature until there is something for it to do. For $\gamma \in \mathcal{M}(s_0)$ put

$$(14) \quad \begin{aligned} \mathcal{N}_1(\gamma) &:= \{\gamma' : d_H(\gamma, \gamma') = 1\}, \\ \mathcal{N}_2(\gamma) &:= \{\gamma' : d_H(\gamma, \gamma') = 2 \text{ and } \exists (k, \ell) \in \gamma \times ([p] \setminus \gamma) \text{ with } \gamma' = \gamma \cup \{\ell\} \setminus \{k\}\}, \end{aligned}$$

and $\mathcal{N}(\gamma) := \mathcal{N}_1(\gamma) \cup \mathcal{N}_2(\gamma)$. The chain proposes, with probability 1/2 each, a single flip of a uniform coordinate of $[p]$ or an exchange of a uniform pair in $\gamma \times ([p] \setminus \gamma)$, and accepts by the Metropolis rule; a proposal outside $\mathcal{M}(s_0)$ is rejected, which is automatic since μ vanishes there. Writing $\tilde{\mathbf{P}}$ for the resulting transition matrix,

$$(15) \quad \tilde{\mathbf{P}}(\gamma, \gamma') = \begin{cases} \frac{1}{2p} \min\left\{1, \frac{\mu(\gamma')}{\mu(\gamma)}\right\}, & \gamma' \in \mathcal{N}_1(\gamma) \cap \mathcal{M}(s_0), \\ \frac{1}{2|\gamma|(p - |\gamma|)} \min\left\{1, \frac{\mu(\gamma')}{\mu(\gamma)}\right\}, & \gamma' \in \mathcal{N}_2(\gamma), \\ 0, & \gamma' \notin \mathcal{N}(\gamma) \cup \{\gamma\}, \end{cases}$$

the diagonal being fixed by $\sum_{\gamma'} \tilde{\mathbf{P}}(\gamma, \gamma') = 1$, and we analyse the lazy chain

$$(16) \quad \mathbf{P} := \frac{1}{2}\mathbf{I} + \frac{1}{2}\tilde{\mathbf{P}}, \quad \mathbf{Q}(\gamma, \gamma') := \mu(\gamma) \mathbf{P}(\gamma, \gamma').$$

LEMMA 3.2 (The kernel). $\mathbf{P}$ is μ -reversible, irreducible and aperiodic on $\mathcal{M}(s_0)$, and non-negative definite, so that its smallest eigenvalue is at least 0 and $\text{Gap}(\mathbf{P}) = 1 - \lambda_2(\mathbf{P})$ is its absolute spectral gap. Moreover $\mathbf{Q}$ is symmetric and, for every $\gamma \in \mathcal{M}(s_0)$ and every $\gamma' \in \mathcal{N}(\gamma) \cap \mathcal{M}(s_0)$,

$$(17) \quad \mathbf{Q}(\gamma, \gamma') \geq \frac{1}{4ps_0} \min\{\mu(\gamma), \mu(\gamma')\}.$$

Convention. From Section 4 onwards the target is $\mu = \pi_n^{(\lambda)}(\cdot | Y)$, and we then write $\tilde{\mathbf{P}}_\lambda$, $\mathbf{P}_\lambda$, $\mathbf{Q}_\lambda$ and $\tau_\epsilon(\mathbf{P}_\lambda)$ for the objects just defined, to record which target they refer to; at $\lambda = 1$ they are the chain of YWJ. Only the acceptance ratio in (15) changes with the temperature. The proposal, the laziness, the underlying graph and every statement of the present subsection and of Section 3.3 do not, and are used below without a subscript wherever the target is immaterial.

3.3. Canonical paths and Sinclair’s bound. The spectral gap of a reversible chain can be bounded below by exhibiting, for each ordered pair of states, a path between them, and checking that no edge of the graph is asked to carry too much probability mass.

Identify $\mathbf{P}$ with the weighted directed graph on $\mathcal{M}(s_0)$ whose edges are the ordered pairs $e = (\gamma, \gamma')$ with $\mathbf{P}(\gamma, \gamma') > 0$, each carrying the weight $\mathbf{Q}(e)$ of (16), which is symmetric by

Lemma 3.2. A *canonical path ensemble* $\mathcal{T}$ assigns to every ordered pair (γ, γ') of distinct states a simple path $T_{\gamma, \gamma'}$ in that graph. Its two characteristics are the *path congestion* and the *maximal length*,

$$(18) \quad \rho(\mathcal{T}) := \max_e \frac{1}{\mathbf{Q}(e)} \sum_{T_{\gamma, \gamma'} \ni e} \mu(\gamma) \mu(\gamma'), \quad \ell(\mathcal{T}) := \max_{\gamma \neq \gamma'} |T_{\gamma, \gamma'}|,$$

the maximum being over directed edges e , and Sinclair's bound (Sinclair, 1992, Corollary 6') reads

$$(19) \quad \text{Gap}(\mathbf{P}) \geq \frac{1}{\rho(\mathcal{T}) \ell(\mathcal{T})}.$$

An arbitrary ensemble is unmanageable, and the device that makes (19) usable is to generate the ensemble from a map. Following YWJ, call $\mathcal{G} : \mathcal{M}(s_0) \rightarrow \mathcal{M}(s_0)$ a *valid transition function* with centre γ^* if $\mathcal{G}(\gamma^*) = \gamma^*$ and, for every $\gamma \neq \gamma^*$, the pair $(\gamma, \mathcal{G}(\gamma))$ is an edge of the graph and $d_H(\mathcal{G}(\gamma), \gamma^*) < d_H(\gamma, \gamma^*)$. The orbit $\gamma \rightarrow \mathcal{G}(\gamma) \rightarrow \mathcal{G}^2(\gamma) \rightarrow \dots$ then reaches γ^* in finitely many steps and is taken as T_{γ, γ^*} ; for a general pair, $T_{\gamma, \gamma'}$ follows T_{γ, γ^*} up to the first state it shares with T_{γ', γ^*} and then follows T_{γ', γ^*} backwards. This is Lemma 1 of YWJ. The ensemble so obtained is *memoryless*: how a path continues depends on the state it has reached and on nothing else.

DEFINITION 3.1 (Precedents; YWJ, Section 4.1). For $\gamma \in \mathcal{M}(s_0)$ put

$$(20) \quad \Lambda(\gamma) := \{\bar{\gamma} \in \mathcal{M}(s_0) : \mathcal{G}^k(\bar{\gamma}) = \gamma \text{ for some } k \geq 0\},$$

the states whose canonical path to γ^* passes through γ ; it contains γ itself, at $k = 0$. For $\bar{\gamma} \in \Lambda(\gamma)$ we write $k(\bar{\gamma})$ for the smallest such k , and $\Lambda_k(\gamma) := \{\bar{\gamma} : k(\bar{\gamma}) = k\}$.

Memorylessness now does all the combinatorial work. A directed edge $e = (\bar{\gamma}, \mathcal{G}(\bar{\gamma}))$ is traversed in the direction of $\mathcal{G}$, so it can occur only in the first segment of a path, which forces the source of that path to be a precedent of $\bar{\gamma}$; this is Lemma A.7c. Since $\sum_{\gamma'} \mu(\gamma') = 1$, the numerator of (18) is at most $\mu[\Lambda(\bar{\gamma})]$; the reverse orientation $(\mathcal{G}(\bar{\gamma}), \bar{\gamma})$ obeys the same bound, by the same argument with the roles of the two endpoints exchanged and because $\mathbf{Q}$ is symmetric. Bounding the denominator by (17),

$$(21) \quad \rho(\mathcal{T}) \leq 4ps_0 \max_{\bar{\gamma} \neq \gamma^*} \max \left\{ 1, \frac{\mu(\bar{\gamma})}{\mu(\mathcal{G}(\bar{\gamma}))} \right\} \cdot \frac{\mu[\Lambda(\bar{\gamma})]}{\mu(\bar{\gamma})}.$$

The length is combinatorial too: each step of $\mathcal{G}$ decreases $d_H(\cdot, \gamma^*)$, so $\ell(\mathcal{T}) \leq 4s_0$, by Lemma A.7b.

Nothing so far has mentioned the temperature, and nothing has used any property of μ . The whole dependence on the target sits in the two factors of (21), and it is there — and only there — that λ will enter. Written out for the target that concerns us, those two factors ask for the two halves of Lemma 4 of YWJ:

- (D $_\lambda$) a *drop bound*: the target does not increase along $\mathcal{G}$, that is $\pi_n^{(\lambda)}(\gamma | Y) \leq \pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)$ for every $\gamma \neq \gamma^*$, and preferably by a definite factor: p^{-2} at an overfitted state and p^{-3} at an underfitted one, as in Lemma 4(a) of YWJ;
- (P $_\lambda$) a *precedent mass bound*: $\pi_n^{(\lambda)}[\Lambda(\gamma) | Y] \leq c \pi_n^{(\lambda)}(\gamma | Y)$ for a constant c and every γ .

Given both, (21) gives $\rho(\mathcal{T}) \leq 4cps_0$ and (19) gives a gap of order $1/(ps_0^2)$. Moreover (P $_\lambda$) is a consequence of (D $_\lambda$) and of counting alone: by telescoping the drop bound along the path,

a precedent at distance k carries at most p^{-3k} times the mass of γ when the states between them are underfitted and at most p^{-2k} when they are overfitted, and by Lemma A.6 there are at most $\{(s^* + 1)p\}^k$ of the first kind and at most p^k of the second; the two geometric series have ratios $(s^* + 1)p^{-2}$ and p^{-1} , and converge. So a single inequality, the drop bound, stands between the assumptions and the spectral gap.

3.4. The transition function. We now recall the map that drives the canonical path ensemble of YWJ. It is defined in terms of the design and of β^* only, and does *not* depend on λ .

DEFINITION 3.2 (States; YWJ, Section 4.1). A state $\gamma \in \mathcal{M}(s_0)$ with $\gamma \neq \gamma^*$ is

- *overfitted* if $\gamma \supseteq \gamma^*$, and *underfitted* otherwise, that is if $\gamma^* \setminus \gamma \neq \emptyset$;
- *saturated* if $|\gamma| = s_0$, and *unsaturated* if $|\gamma| < s_0$.

An overfitted $\gamma \neq \gamma^*$ satisfies $\gamma \setminus \gamma^* \neq \emptyset$, and an underfitted state can only exist if $s^* \geq 1$.

DEFINITION 3.3 (Greedy transition function; YWJ, Section 4.1). For $\gamma \in \mathcal{M}(s_0) \setminus \{\gamma^*\}$ define $\mathcal{G}(\gamma)$ by the first applicable rule:

(G1) γ *overfitted*. Let

$$(22) \quad \ell_\gamma \in \arg \min_{\ell \in \gamma \setminus \gamma^*} \left\{ \|\Phi_\gamma X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 - \|\Phi_{\gamma \setminus \{\ell\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 \right\},$$

the least influential uninfluential covariate present, and set $\mathcal{G}(\gamma) := \gamma \setminus \{\ell_\gamma\}$.

(G2) γ *underfitted and unsaturated*. Let

$$(23) \quad j_\gamma \in \arg \max_{j \in \gamma^* \setminus \gamma} \|\Phi_{\gamma \cup \{j\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2,$$

the missing influential covariate explaining most signal, and set $\mathcal{G}(\gamma) := \gamma \cup \{j_\gamma\}$.

(G3) γ *underfitted and saturated*. With j_γ as in (23) and $\tilde{\gamma} := \gamma \cup \{j_\gamma\}$, let

$$(24) \quad k_\gamma \in \arg \min_{k \in \tilde{\gamma} \setminus \gamma^*} \left\{ \|\Phi_{\tilde{\gamma}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 - \|\Phi_{\tilde{\gamma} \setminus \{k\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 \right\},$$

and set $\mathcal{G}(\gamma) := \tilde{\gamma} \setminus \{k_\gamma\} = \gamma \cup \{j_\gamma\} \setminus \{k_\gamma\}$.

Ties in (22)–(24) are broken by the lexicographic order, and $\mathcal{G}(\gamma^*) := \gamma^*$.

Rules (G1) and (G2) decrease $d_H(\cdot, \gamma^*)$ by one and rule (G3) decreases it by two, so $\mathcal{G}$ is a valid transition function in the sense of YWJ's Lemma 1 and generates a memoryless canonical path ensemble directed at γ^* , of maximal path length at most $2s_0$.

We record the three size changes once and for all, since they are what (8) feeds on:

$$(25) \quad |\mathcal{G}(\gamma)| - |\gamma| = \begin{cases} -1 & \text{under rule (G1),} \\ +1 & \text{under rule (G2),} \\ 0 & \text{under rule (G3).} \end{cases}$$

3.5. From the spectral gap to the mixing time. A spectral gap is not yet a mixing time. For the lazy chain $\mathbf{P}$ of Section 3.2, or indeed any lazy reversible chain with stationary law μ on a finite state space, the ϵ -mixing time $\tau_\epsilon := \max_\gamma \min\{t : \|\mathbf{P}^{t'}(\gamma, \cdot) - \mu\|_{\text{TV}} \leq \epsilon \forall t' \geq t\}$ obeys

$$(26) \quad \tau_\epsilon \leq \frac{\log[1/\min_\gamma \mu(\gamma)] + \log(1/\epsilon)}{\text{Gap}(\mathbf{P})},$$

which is the upper half of the sandwich used by YWJ in their equation (14); see [Sinclair \(1992\)](#).

The numerator of (26) is where posterior concentration re-enters, and it enters modestly. YWJ bound $\min_{\gamma} \pi_n(\gamma | Y)$ below by factorising $\pi_n(\gamma | Y) = \pi_n(\gamma^* | Y) \cdot \{\pi_n(\gamma | Y) / \pi_n(\gamma^* | Y)\}$: the first factor is bounded below by Theorem 3.1 and the second by the explicit ratio (8) at $\lambda = 1$, whose worst case costs a factor $p^{-\kappa s_0} (1 + g)^{-(n+s_0)/2}$ — this is Lemma 5.2 at $\lambda = 1$. Taking logarithms, the numerator is $\mathcal{O}\{(\alpha n + \alpha s_0 + \kappa s_0) \log p + \log(1/\epsilon)\}$, and their gap bound $\text{Gap}(\mathbf{P}) \geq 1/(24ps_0^2)$ turns this into their Theorem 2,

$$\tau_{\epsilon} \leq 12ps_0^2\{(\alpha n + \alpha s_0 + 2\kappa s_0) \log p + \log(1/\epsilon) + \log 2\}.$$

Two features of this last step are worth isolating, because both survive tempering unchanged. The concentration statement is used only through the smallest mass, so any constant lower bound on $\pi_n(\gamma^* | Y)$ would serve equally well: the $(n + s_0) \log p$ term comes from the ratio, not from the quality of the concentration. And that term is a burn-in cost; started at γ^* rather than at the worst state, the chain does not pay it.

4. The drop bound at temperature λ . We come to (D_{λ}) , for $\pi_n^{(\lambda)}(\cdot | Y)$ along the map $\mathcal{G}$ of Definition 3.3. This section establishes it, together with its consequence (P_{λ}) , above an explicit threshold, and shows that no such threshold can be dispensed with.

THEOREM 4.1 (Drop bound at temperature λ). *Assume Assumptions 3.1 to 3.5, and work on the good event. Let $\lambda \geq 0$ and let $\gamma \in \mathcal{M}(s_0)$ with $\gamma \neq \gamma^*$. Then, if γ is overfitted,*

$$(27) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq -\kappa \log p - \lambda \{\alpha - 8(L + \tilde{L})\} \log p;$$

if γ is underfitted and unsaturated,

$$(28) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq \kappa \log p + \lambda \left\{ \alpha \log p - \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 \bar{\beta}^2 (\gamma^* \setminus \gamma)}{64\sigma_0^2} \right\} \right\};$$

and if γ is underfitted and saturated,

$$(29) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq -\frac{\lambda}{2} \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 \bar{\beta}^2 (\gamma^* \setminus \gamma)}{64\sigma_0^2} \right\}.$$

The proof is deferred to Sections A.4 to A.6. It is three applications of (8), one per rule of Definition 3.3, with the size change read off from (25); the estimates it uses are the deterministic bounds of Section A, none of which involves λ .

Read the three bounds as functions of λ . Each is affine, and the three intercepts and the three slopes tell the whole story.

- (i) (27) has intercept $-\kappa \log p$ and slope $-\{\alpha - 8(L + \tilde{L})\} \log p$, an identity now that (11) fixes α . It uses no condition on λ whatsoever, and when $\alpha > 8(L + \tilde{L})$ its worst value on $[0, 1]$ is at $\lambda = 0$, where it reads $-\kappa \log p$. Deleting an uninformative covariate is favourable at every temperature, and least favourable at the prior, where the sparsity exponent κ alone does the work.
- (ii) (28) has intercept $+\kappa \log p$ and slope $\alpha \log p - \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_{\beta}^2}{64\sigma_0^2} \right\}$. The intercept is positive: at $\lambda = 0$ the bound is useless, and Proposition 4.1 shows that this is not a defect of the bound. The slope is negative precisely when the greedy gain exceeds the Occam factor of the g -prior, and then the bound becomes useful above the temperature at which the two cross.

(iii) (29) has no intercept at all: the inclusion penalty cancels identically in an exchange, both models having the same size and the prior depending on the model only through its size, so that $\pi_n^{(0)}(\gamma | Y) = \pi_n^{(0)}(\gamma' | Y)$ exactly. Exchange is neutral under the prior and favourable as soon as the likelihood is switched on; the temperature it needs to reach the exponent p^{-3} differs from the one case (ii) needs by the factor $\{6/(\kappa + 3)\}\{1 - \alpha \log p / \min\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\}\}$, so it is the smaller of the two whenever $\kappa \geq 3$, and can be the larger only when $\kappa < 3$ and the Occam discount in the second brace is close to 1. Which of them determines λ_* in (31) therefore depends on the design.

Behind the three cases is the tension already visible in (8): the untempered prior charges $\kappa \log p$ per covariate at every temperature, while the fit term is damped by λ . The map $\mathcal{G}$ knows nothing of it, being built from X and β^* alone and directed at γ^* whatever the temperature. At $\lambda = 1$ that is the right destination; at $\lambda = 0$ the target is the prior, whose mode is the null model, and each of the s^* additions $\mathcal{G}$ must make divides the target by exactly p^κ . So no such map can satisfy (D $_\lambda$) at every temperature. What is true, and is what this section proves, is that the failure occupies an initial segment and nothing more: there is an explicit λ_* above which the argument of Section 3 transposes with no change beyond the substitution of $\pi_n^{(\lambda)}$ for π_n .

4.1. *The critical temperature.* We ask for the conclusion in the form Lemma 4(a) of YWJ gives it — each ratio at most p^{-2} at an overfitted state and at most p^{-3} at an underfitted one — rather than merely “at most one”, because Section 3.3 needs a definite factor to sum the precedents against, and because those are the two exponents the two cases can afford.

COROLLARY 4.1 (Drop bound at temperature λ ; YWJ, Lemma 4(a)). *Assume the hypotheses of Theorem 4.1, and write $\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\}$ for the floor of Theorem 4.1 at its worst, that is with $\overline{\beta^2}(\gamma^* \setminus \gamma)$ replaced by its lower bound C_β^2 from (4).*

(a) *For every $\lambda \in [0, 1]$ and every overfitted $\gamma \in \mathcal{M}(s_0) \setminus \{\gamma^*\}$,*

$$(30) \quad \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq p^{-\{(1-\lambda)\kappa + \lambda[\kappa + \alpha - 8(L + \tilde{L})]\}} \leq p^{-2}.$$

(b) *Suppose $\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\} > \alpha \log p$ and set*

$$(31) \quad \lambda_* := \max \left\{ \frac{(\kappa + 3) \log p}{\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\} - \alpha \log p}, \quad \frac{6 \log p}{\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\}} \right\}.$$

Then, for every $\lambda \geq \lambda_$ and every underfitted $\gamma \in \mathcal{M}(s_0)$,*

$$(32) \quad \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq p^{-3}.$$

At $\lambda = 1$ this is Lemma 4(a) of YWJ, exponents included. The two halves behave quite differently in λ , and (30) says why the overfitted one needs no threshold: by (8) the log-ratio is affine in λ , hence is the interpolation

$$(33) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} = (1 - \lambda) \log \frac{\pi(\gamma)}{\pi(\mathcal{G}(\gamma))} + \lambda \log \frac{\pi_n(\gamma | Y)}{\pi_n(\mathcal{G}(\gamma) | Y)}$$

between the prior ratio and the posterior ratio, and at an overfitted state *both* endpoints are favourable: the first is exactly $-\kappa \log p$ and the second is at most $-\{\kappa + \alpha - 8(L + \tilde{L})\} \log p$ by (27), so the whole segment lies below the smaller of the two, which Assumption 3.3 puts at $-2 \log p$. Deleting an uninfluential covariate is favoured by the sparsity prior and by the likelihood alike — the Occam factor $(1 + g)^{-1/2}$ charges for the extra parameter, and a covariate outside γ^* buys only noise — so there is nothing for a temperature to arbitrate. At an underfitted state the endpoints no longer agree. For an addition the prior one is $+\kappa \log p$, of the opposite sign, and the segment reaches $-3 \log p$ only at the first term of (31); for an exchange it is exactly 0, the inclusion penalty cancelling, and the crossing is at the second. Above the larger of the two, which is λ_* , both are past it, and Proposition 4.1 shows that below it the first genuinely is not.

Part b is a conditional statement, and says nothing until one knows that λ_* is below 1. That is where the two hypotheses of Section 3 which do the arithmetic enter, one per term of the minimum defining the floor.

COROLLARY 4.2 (The critical temperature is below one). *Assume Assumptions 3.1 to 3.6. Then*

$$(34) \quad \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} \geq 2(L + \tilde{L} + \alpha + \kappa) \log p > \alpha \log p,$$

so Corollary 4.1b applies, and

$$(35) \quad \lambda_* \leq 1.$$

The proof gives more than it is asked for. The two hypotheses that make the floor large, Assumptions 3.5 and 3.6, are each exactly the statement $\min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} \geq 2(L + \tilde{L} + \alpha + \kappa) \log p$, one per term of the minimum; and once the floor is measured in those units, every quantity in (31) is a ratio of the numerical constants of Section 3. The design, the sample size, the dimension and the signal strength cancel.

COROLLARY 4.3 (The critical temperature is below one half). *Assume Assumptions 3.1 to 3.6. Then*

$$(36) \quad \lambda_* < \frac{1}{2},$$

and $\frac{1}{2}$ is the exact supremum of the right-hand side of (31) over the parameters $(L, \tilde{L}, \alpha, \kappa)$ that Assumptions 3.2 and 3.3 admit; it is approached, and not attained, as $\kappa \rightarrow \infty$ with $L + \tilde{L} = 4$ and $\alpha = \frac{1}{2}$. The same holds when the exponent 3 of (32) is replaced by any $q \leq 4$, so that (31) becomes

$$(37) \quad \lambda_*(q) := \max \left\{ \frac{(\kappa + q) \log p}{\min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} - \alpha \log p}, \frac{2q \log p}{\min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\}} \right\}.$$

Three consequences are worth separating.

- (i) *The region left uncovered by Section 5 is a fixed initial segment.* Everything from Corollary 4.1 to Theorem 5.1 holds on $[\lambda_*, 1] \supseteq [\frac{1}{2}, 1]$, an interval of length at least one half whatever the design. Whatever Section 6 must do, it must do on $[0, \frac{1}{2}]$, where the likelihood is damped by a factor of at least two.

- (ii) *The exponent 3 is not what makes this work.* By (37) the exponent enters λ_* only through $(\kappa + q)$ and $2q$, and the supremum $\frac{1}{2}$ is the same for every $q \leq 4$. Lowering it to 2 — which Corollary 4.4 permits, since the underfitted series there needs only $(s^* + 1)p^{-1} \leq \frac{1}{2}$, and Assumption 3.4 gives $s^* + 1 \leq s \leq n/(256 \log p)$ — multiplies λ_* by $(\kappa + 2)/(\kappa + 3)$ and moves nothing else; the concentration of (43) would then read $1 - \mathcal{O}(s^*p^{-1})$ in place of $1 - \mathcal{O}(p^{-1})$. The third power buys the rate of Proposition 5.1, not the threshold.
- (iii) *It does not, by itself, close the path.* A bound valid below λ_* is still needed, and that is what Section 6 supplies — at every temperature, as it turns out, so that the threshold disappears from the final statement.

4.2. *The mass of the precedents.* Bound (P_λ) transposes with the two exponents that Corollary 4.1 supplies, and the counting is again theirs.

COROLLARY 4.4 (Mass of the precedents at temperature λ ; YWJ, Lemma 4(b)). *Let $\lambda \in [\lambda_*, 1]$ satisfy the hypotheses of Corollary 4.1, and let $p \geq 3$. Then, for every $\gamma \in \mathcal{M}(s_0)$,*

$$(38) \quad \frac{\pi_n^{(\lambda)}[\Lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \frac{1}{\{1 - (s^* + 1)p^{-2}\}\{1 - p^{-1}\}} \leq 4.$$

The two ratios summed here are $a := (s^* + 1)p^{-2}$, over the underfitted precedents, where each step costs p^{-3} and there are at most $(s^* + 1)p$ choices, and $b := p^{-1}$, over the overfitted ones, where each step costs p^{-2} and there are at most p choices. Since $s^* \leq p$, both are at most $\frac{1}{2}$ for every $p \geq 3$: unlike the condition $(s^* + 1)p^{-1} \leq \frac{1}{2}$ that a single common exponent 2 would impose, YWJ's pair asks nothing of s^* relative to p . The constant 4 is $\{(1 - a)(1 - b)\}^{-1}$ and tends to 1 with p ; theirs is 6.

4.3. *The restriction on λ is real.* Item (ii) above asserted that the positive intercept of (28) belongs to the problem and not to the estimate. The following identity, valid for every design and every realisation of the data, is that assertion.

PROPOSITION 4.1 (Exact threshold). *Let $\gamma \in \mathcal{M}(s_0)$ be underfitted and unsaturated, so that $\mathcal{G}(\gamma) = \gamma \cup \{j_\gamma\}$. Then, for every $\lambda \geq 0$,*

$$(39) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} = \kappa \log p - \lambda \log \frac{\mathcal{L}_n(Y | \mathcal{G}(\gamma))}{\mathcal{L}_n(Y | \gamma)}.$$

Defining

$$(40) \quad \lambda_{\text{crit}} := \max \left\{ \frac{\kappa \log p}{\log \{ \mathcal{L}_n(Y | \mathcal{G}(\gamma)) / \mathcal{L}_n(Y | \gamma) \}} : \right. \\ \left. \gamma \in \mathcal{M}(s_0) \text{ underfitted and unsaturated} \right\} \in (0, \infty],$$

the map $\mathcal{G}$ is non-decreasing for $\pi_n^{(\lambda)}(\cdot | Y)$ at every underfitted unsaturated state if and only if $\lambda \geq \lambda_{\text{crit}}$. Finally, on the good event and under the hypotheses of Theorem 4.1,

$$(41) \quad \lambda_{\text{crit}} \leq \frac{\kappa \log p}{\min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} - \alpha \log p} = \lambda_*(0)$$

whenever $\min \left\{ \frac{n\nu}{16s^}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} > \alpha \log p$.*

5. The spectral gap and the mixing time above λ_* . Corollaries 4.1 and 4.4 supply (D_λ) and (P_λ) above λ_* , so the machinery of Section 3 runs unchanged. Nothing in this section is new relative to that one, and that is the point.

5.1. The ensemble and its congestion. Let $\mathcal{T}$ be the canonical path ensemble generated by $\mathcal{G}$ in the sense of Section 3.3: $\mathcal{G}$ is a valid transition function with centre γ^* , by the second remark after Definition 3.3, so T_{γ, γ^*} is the orbit $\gamma \rightarrow \mathcal{G}(\gamma) \rightarrow \dots \rightarrow \gamma^*$ and a general $T_{\gamma, \gamma'}$ is obtained from two such orbits. Every edge it uses joins some $\bar{\gamma}$ to $\mathcal{G}(\bar{\gamma})$, hence lies in $\mathcal{N}(\bar{\gamma})$ and carries positive weight by Lemma 3.2.

LEMMA 5.1 (Spectral gap at temperature λ). *Let λ satisfy the hypotheses of Corollary 4.4. Then, on the good event,*

$$(42) \quad \rho(\mathcal{T}) \leq 16 p s_0 \quad \text{and} \quad \text{Gap}(\mathbf{P}_\lambda) \geq \frac{1}{64 p s_0^2}.$$

Two features are worth recording. The bound does not involve the exponents of Corollary 4.1: those fix only the interval $[\lambda_*, 1]$ of temperatures on which it applies, an interval that Corollary 4.2 makes non-empty. And it does not involve λ either, so the same rate $1/(64 p s_0^2)$ holds simultaneously at every temperature in that interval — which is what an SMC complexity bound needs, since it requires a gap at each intermediate distribution and not only at the last.

5.2. Concentration at temperature λ . No statement of the form of Theorem 3.1 can hold at every $\lambda < 1$: at $\lambda = 0$ the target is the sparsity prior, which places mass of order $p^{-\kappa s^*}$ on γ^* . Above λ_* , however, concentration comes for free from Corollary 4.4, because γ^* is the state all canonical paths end at.

PROPOSITION 5.1 (Concentration at temperature λ). *Under the hypotheses of Corollary 4.4, on the good event,*

$$(43) \quad \pi_n^{(\lambda)}(\gamma^* | Y) \geq \{1 - (s^* + 1)p^{-2}\} \{1 - p^{-1}\} \geq \frac{1}{4}.$$

The bound (43) is $1 - \mathcal{O}(p^{-1})$, which is the rate of Theorem 3.1; a common exponent 2 in Corollary 4.1 would have given only $1 - \mathcal{O}(s^* p^{-1})$, and this is the one place where YWJ's third power earns something. Unlike Theorem 3.1 the bound is deterministic on the good event and holds at every $\lambda \geq \lambda_*$, and it needs no separate argument: concentration and rapid mixing are here two readings of the same drop bound.

5.3. The smallest mass and the mixing time. Only one ingredient of the sandwich (26) is still missing, the smallest mass, and Proposition 5.1 supplies it: a target that puts at least a quarter of its mass on γ^* cannot put too little anywhere, because (8) bounds every ratio to γ^* explicitly.

LEMMA 5.2 (Lower bound on the smallest mass). *For every $\lambda \in [0, 1]$ and every $\gamma \in \mathcal{M}(s_0)$,*

$$(44) \quad \log \frac{\pi_n^{(\lambda)}(\gamma^* | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \kappa s_0 \log p + \lambda(n + s_0) \alpha \log p.$$

Consequently, under the hypotheses of Corollary 4.4,

$$(45) \quad \min_{\gamma \in \mathcal{M}(s_0)} \pi_n^{(\lambda)}(\gamma | Y) \geq \frac{1}{4} p^{-\kappa s_0} (1 + g)^{-\lambda(n + s_0)/2}.$$

Feeding Lemmas 5.1 and 5.2 into (26) gives the tempered form of Theorem 2 of Yang, Wainwright and Jordan (2016).

THEOREM 5.1 (Mixing time at temperature λ). *Let λ satisfy the hypotheses of Corollary 4.4. Then, on the good event, for every $\epsilon \in (0, 1)$,*

$$(46) \quad \tau_\epsilon(\mathbf{P}_\lambda) \leq 64 p s_0^2 \left\{ \kappa s_0 \log p + \lambda (n + s_0) \alpha \log p + \log \frac{1}{\epsilon} + \log 4 \right\}.$$

6. A gap at every temperature. Proposition 4.1 says that no ensemble directed at γ^* can work below λ_* : at $\lambda = 0$ the ratio along $\mathcal{G}$ is exactly p^κ at every underfitted unsaturated state. Corollary 4.3 says how much is at stake, namely the fixed initial segment $[0, \frac{1}{2}]$. And (9) says that nothing worse is at stake: the two directions $\gamma \rightarrow \gamma \cup \{j\}$ and $\gamma \cup \{j\} \rightarrow \gamma$ have log-ratios that are exact negatives, so at every state and every coordinate one of them is uphill. What fails below λ_* is not the existence of an uphill move but the assertion that it is the addition.

We therefore keep Sinclair's bound and change the destination. The centre of the ensemble is a subset of γ^* that the temperature selects, growing from $\emptyset$ at $\lambda = 0$ to γ^* at $\lambda = 1$; the states outside γ^* are funnelled away first, by a map that no temperature can turn around. What makes such a centre computable is that the whole tempering path is a two-parameter exponential family in which λ does nothing but rotate a tilt.

6.1. *The target in the residual variable.* Write

$$(47) \quad D_\gamma := g^{-1} \|Y\|_2^2 + \|(I - \Phi_\gamma)Y\|_2^2 = \|Y\|_2^2 (g^{-1} + 1 - R_\gamma^2)$$

for the regularised residual sum of squares, so that $1 + g(1 - R_\gamma^2) = g D_\gamma / \|Y\|_2^2$.

LEMMA 6.1 (The tempering path is an exponential family). *For every $\gamma \in \mathcal{M}(s_0)$ and every $\lambda \geq 0$,*

$$(48) \quad \pi_n^{(\lambda)}(\gamma | Y) = \frac{w_\lambda(\gamma)}{\sum_{\gamma' \in \mathcal{M}(s_0)} w_\lambda(\gamma')}, \quad w_\lambda(\gamma) := \exp \left\{ -\frac{\lambda n}{2} \log D_\gamma - (\kappa + \lambda \alpha) |\gamma| \log p \right\}.$$

This is an identity, not an estimate, and it is (8) written once and for all: the path is the exponential family with sufficient statistic $(\log D_\gamma, |\gamma|)$ and natural parameter $(-\lambda n/2, -(\kappa + \lambda \alpha) \log p)$. Raising the temperature rotates that parameter from *prefer few covariates* towards *prefer a small residual*, and changes nothing else about the target. It is why a single family of routes can serve every temperature, and why the only quantity below that depends on λ is the point of the rotation at which the tilt balances.

For $\gamma \in \mathcal{M}(s_0)$ with $|\gamma| < s_0$ and $j \in [p] \setminus \gamma$ put, as in YWJ's notation,

$$(49) \quad \psi_j(\gamma) := \log \frac{\mathcal{L}_n(Y | \gamma \cup \{j\})}{\mathcal{L}_n(Y | \gamma)} = -\alpha \log p + \frac{n}{2} \log \frac{D_\gamma}{D_{\gamma \cup \{j\}}},$$

the second equality by (6) and (11), so that (8) with $|\gamma'| - |\gamma| = 1$ reads

$$(50) \quad \log \frac{\pi_n^{(\lambda)}(\gamma \cup \{j\} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} = \lambda \psi_j(\gamma) - \kappa \log p,$$

which is (9).

6.2. *Fibres, and the factorisation.* Theorem 4.1 bounds ψ_j from below, which is what an addition needs; the argument below needs a bound from above, and for the uninfluential covariates only. That is the one hypothesis of this section which is not already in Section 3.

ASSUMPTION 6.1 (Log-gains). There is $a_0 < \infty$ such that, on the good event, $\psi_j(\gamma) \leq (a_0 - \alpha) \log p$ for every $\gamma \in \mathcal{M}(s_0)$ with $|\gamma| < s_0$ and every $j \in S^c \setminus \gamma$. We assume $\kappa + \alpha \geq 2 + a_0$, which with $\kappa \geq 2$ from Assumption 3.3 gives

$$(51) \quad \kappa_\lambda := \kappa - \lambda(a_0 - \alpha) \geq 2 \quad \text{for every } \lambda \in [0, 1].$$

Read (51) through (50): it says that for every uninfluential j and every context,

$$(52) \quad \pi_n^{(\lambda)}[\gamma_j = 1 \mid \gamma_{-j}] \leq p^{-\kappa_\lambda} \leq p^{-2},$$

the conditional vanishing outright when $|\gamma| = s_0$. The uninfluential coordinates are *pinned*: each is a nearly deterministic zero whatever the others do. That is a statement about dependence, not about routes, and it is what the whole section rests on.

DEFINITION 6.1 (Fibres). For $\sigma \subseteq S^c$ with $|\sigma| \leq s_0$ put

$$(53) \quad F_\sigma := \{\gamma \in \mathcal{M}(s_0) : \gamma \setminus S = \sigma\},$$

so that the F_σ partition $\mathcal{M}(s_0)$, each is a copy of $\{A \subseteq S : |A| \leq s_0 - |\sigma|\}$, and each carries at most 2^{s^*} states. Write μ_σ for $\pi_n^{(\lambda)}(\cdot \mid F_\sigma)$ and $\mathbf{P}_\lambda|_{F_\sigma}$ for $\mathbf{P}_\lambda$ restricted to F_σ with rejection, and put

$$(54) \quad \kappa_S(\lambda) := \min_{\sigma: |F_\sigma| \geq 2} \text{Gap}(\mathbf{P}_\lambda|_{F_\sigma}).$$

A fibre fixes every uninfluential coordinate and lets the influential ones move; $\kappa_S(\lambda)$ is the worst rate at which the chain equilibrates the influential block with the rest held fixed. It is a gap on at most 2^{s^*} states, and it is the only place where s^* will occur.

LEMMA 6.2 (Approximate block factorisation). Let $\lambda \in [0, 1]$, assume Assumption 6.1, let $p \geq 9$, and work on the good event. Then, for every $f : \mathcal{M}(s_0) \rightarrow \mathbb{R}$,

$$(55) \quad \text{Var}_{\pi_n^{(\lambda)}}(f) \leq C_p \left[\sum_{j \in S^c} \mathbb{E} \text{Var}(f \mid \gamma_{-j}) + \mathbb{E} \text{Var}(f \mid \gamma_{S^c}) \right], \quad C_p := (1 - \sqrt{2/p})^{-1},$$

the conditional variances being taken under $\pi_n^{(\lambda)}(\cdot \mid Y)$ and the expectations over the conditioning variables.

The blocks of (55) are the influential set S , taken whole, and the uninfluential coordinates, taken one at a time. The constant is absolute: it does not involve s_0 , s^* , n or the design, only the margin that (52) provides. What makes it absolute is that the dependence between the blocks runs in one direction only. A null coordinate is barely influenced by anything, by (52); the influential block may well be strongly influenced by a null coordinate, but only through an event of probability p^{-2} , so the round trip $S \rightarrow \{j\} \rightarrow S$ has gain at most $2/p$ summed over the p coordinates, and it is the round trip that a Dobrushin argument controls.

6.3. The gap.

THEOREM 6.1 (A gap at every temperature). *Let $\lambda \in [0, 1]$, $p \geq 9$, assume Assumption 6.1, and work on the good event. Then*

$$(56) \quad \text{Gap}(\mathbf{P}_\lambda) \geq \frac{1}{C_p \{4p + 1/\kappa_S(\lambda)\}} \geq \frac{1}{2C_p} \min\left\{\frac{1}{4p}, \kappa_S(\lambda)\right\}.$$

There is no s_0 in (56), and no s^* . The two terms are the two blocks of (55): $4p$ is the rate at which the chain proposes a given coordinate, and $1/\kappa_S(\lambda)$ is the cost of the influential block. Three remarks on how this differs from a path argument, and on what it does not remove.

REMARK 6.1 (Why the path method loses s_0 , and this does not). A canonical path ensemble must connect a state carrying k uninfluential covariates to one carrying none, and any such path has at least k edges; Sinclair's bound charges the longest one. That charge is not an artefact of a bad choice of ensemble. For the funnel of the previous version, the demand across an edge at depth k is at most $e^{1/p}$ times the mass of its own endpoint while $\mathbf{Q}_\lambda$ of that edge is that mass over $4p$, so the ratio is $4pe^{1/p}$ at every depth; in the weighted form of Diaconis and Stroock (1991) the bound is then at least $4p(\max_k w_k)(\sum_{k \leq s_0} 1/w_k) \geq 4ps_0$, whatever the weights w . Factorisation charges no length at all: each uninfluential coordinate is equilibrated on its own, at rate $1/(4p)$, and s_0 never appears.

REMARK 6.2 (κ_S is necessary). No hypothesis-free improvement of (56) is possible, because the gap itself is not bounded below by a rational function of p, s_0, s^* . Take two collinear columns with cancelling coefficients, $\beta^* = (b, -b, 0, \dots, 0)$ with X_0, X_1 correlated at r close to one: neither explains anything alone and the pair explains everything, so at the temperature at which the two heavy configurations of the influential block are comparable both singletons are crushed, and every path between them — for the chain as much as for any ensemble — must cross one of them. That is a bottleneck *inside a fibre*, so it is $\kappa_S(\lambda)$ that must record it, and Section 7 shows that it does, to within a factor 2.1 over three orders of magnitude.

REMARK 6.3 (What is left to bound). (56) reduces the problem to a chain on at most 2^{s^*} states. That chain inherits the proposal rate of the original, so $\kappa_S(\lambda) = \mathcal{O}(1/p)$ and the two terms of (56) are of the same order; the content of the theorem is that s_0 has left and that what remains is a question about the influential block alone. Section 6.5 bounds $\kappa_S(\lambda)$ by a canonical path argument *on the fibre*, where the path length is at most $2s^*$ and the conclusion is

$$(57) \quad \kappa_S(\lambda) \geq \frac{1}{8p(s^*)^2 \min\{C_\lambda, M_\lambda\}},$$

so that (56) becomes $\text{Gap}(\mathbf{P}_\lambda) \geq c/\{p(s^*)^2 \min\{C_\lambda, M_\lambda\}\}$ — the same shape as before with s_0 replaced by s^* .

6.4. The greedy chain and the ascent. It remains to join the 2^{s^*} pure states. Both moves of (14) are available between subsets of γ^* , and the route uses both: exchanges to reach the greedy chain, flips to travel along it.

DEFINITION 6.2 (The greedy chain and the size the temperature selects). Let $P_0 := \emptyset$ and, for $0 \leq m < s^*$,

$$(58) \quad j_{m+1} \in \arg \min_{j \in \gamma^* \setminus P_m} D_{P_m \cup \{j\}}, \quad P_{m+1} := P_m \cup \{j_{m+1}\},$$

ties broken lexicographically, so that $P_{s^*} = \gamma^*$. Put

$$(59) \quad g_m := \frac{n}{2} \log \frac{D_{P_m}}{D_{P_{m+1}}} - \alpha \log p = \psi_{j_{m+1}}(P_m), \quad \varphi_\lambda(m) := \log w_\lambda(P_m),$$

so that $\varphi_\lambda(m+1) - \varphi_\lambda(m) = \lambda g_m - \kappa \log p$, and

$$(60) \quad m_\lambda := \min_{0 \leq m \leq s^*} \arg \max \varphi_\lambda(m).$$

The gains g_m are the marginal log-gains of the greedy, that is best-subset, regression path on the influential set; they are functions of (X, Y, g) and of nothing else. The single quantity in this section that depends on the temperature is m_λ , the size the tilt selects, and $m_0 = 0$ while $m_1 = s^*$ as soon as $\lambda_* \leq 1$.

The destination is P_{m_λ} , and what remains is to choose the way there. Two requirements pull against each other: the route must be *short*, which asks for a fixed itinerary, and it must be *uphill*, which asks for freedom to choose. We meet both by fixing a potential and taking the best move that decreases it.

DEFINITION 6.3 (The potential and the ascent). For $A \subseteq \gamma^*$ put

$$(61) \quad V(A) := |A \setminus P_{|A|}| + ||A| - m_\lambda|,$$

and call a neighbour of A in $(2^{\gamma^*}, \text{flips and exchanges})$ *admissible* if it has a strictly smaller V . Let

$$(62) \quad F_\lambda(A) := \arg \max \{w_\lambda(z) : z \text{ an admissible neighbour of } A\}, \quad F_\lambda(P_{m_\lambda}) := P_{m_\lambda},$$

ties broken lexicographically. The second-level route $R_{A,A'}$ is the F_λ -orbit of A followed by the reverse of that of A' , truncated at their first common state; and for $x \in B_A$, $y \in B_{A'}$ the canonical path $\tilde{T}_{x,y}$ is the F -orbit of x , then $R_{A,A'}$, then the reverse of the F -orbit of y . For $A = A'$ it is the path built from F inside B_A as in Section 5.1.

Two admissible moves always exist, and they are the two segments of the obvious itinerary — *align*, then *climb*:

- (E1) *align*. If $A \neq P_{|A|}$, write $A \setminus P_{|A|} = \{a_1 < \dots < a_q\}$ and $P_{|A|} \setminus A = \{b_1 < \dots < b_q\}$, which have the same cardinality, and exchange a_1 for b_1 . This is an $\mathcal{N}_2$ move and it decreases the first term of (61);
- (E2) *climb or descend*. If $A = P_l$, add j_{l+1} when $l < m_\lambda$ and delete j_l when $l > m_\lambda$. This is an $\mathcal{N}_1$ move and it decreases the second term.

So F_λ is defined at every $A \neq P_{m_\lambda}$, and $V(A) = 0$ exactly at $A = P_{m_\lambda}$. Since V drops by at least one at each step and $V \leq 2s^*$, every orbit has at most $2s^*$ edges and the longest route satisfies

$$(63) \quad L_\lambda \leq 4s^*.$$

The exchange is what makes (E1) possible at all: two competing collinear covariates put almost all of $\bar{\pi}_\lambda$ on the two singletons and almost none on $\emptyset$, and a route that flips them one at a time passes through exactly that hole, whereas an exchange steps between them directly.

The ascent has one weakness, and it is the mirror image of that strength. When many states carry comparable mass the choice in (62) is arbitrary, and an arbitrary choice sends all the traffic one way; Remark 6.6 makes that precise. Against it we put a second route, which ignores the mass entirely.

DEFINITION 6.4 (The crossover route). Order γ^* by the greedy chain, $j_1, \dots, j_{s^*}$, and route from A to A' by setting the coordinates to their A' -values in that order: the t -th state of the route is

$$(64) \quad z_t := (A' \cap \{j_1, \dots, j_t\}) \cup (A \cap \{j_{t+1}, \dots, j_{s^*}\}),$$

a coordinate being skipped when A and A' agree on it. Every step is a flip, the route has at most s^* of them, and we write $\varrho_\lambda^\times$ for its congestion in the sense of (65).

DEFINITION 6.5 (Congestion, cut ratio, product defect). Let f be any flow on $(2^{\gamma^*}, \text{flips and exchanges})$: any way of routing a demand $\bar{\pi}_\lambda(A)\bar{\pi}_\lambda(A')$ between each ordered pair along one or several paths, a single family $\{R_{A,A'}\}$ being the special case that puts all of each demand on one route. Its *congestion* and longest route are

$$(65) \quad \varrho_\lambda := \max_{e=(z,z')} \frac{1}{\min\{\bar{\pi}_\lambda(z), \bar{\pi}_\lambda(z')\}} \sum_{R_{A,A'} \ni e} \bar{\pi}_\lambda(A) \bar{\pi}_\lambda(A'), \quad L_\lambda,$$

the maximum being over directed edges. For the ascent of Definition 6.3, write $\Lambda^{F_\lambda}(z)$ for the set of A whose orbit passes through z and

$$(66) \quad x_\lambda(z) := \bar{\pi}_\lambda[\Lambda^{F_\lambda}(z)], \quad C_\lambda := \max_{z \neq P_{m_\lambda}} \frac{x_\lambda(z) \{1 - x_\lambda(z)\}}{\bar{\pi}_\lambda(z)},$$

the *cut ratio*. For the crossover, order γ^* by the greedy chain and write, for $0 \leq t \leq s^*$, $I_{<t} := \{j_1, \dots, j_{t-1}\}$ and $I_{\geq t}$ for its complement; then

$$(67) \quad M_\lambda := \max_{0 \leq t \leq s^*} \max_{a \in I_{<t}, b \in I_{\geq t}} \frac{\bar{\pi}_\lambda(a) \bar{\pi}_\lambda(b)}{\bar{\pi}_\lambda(a \cup b)},$$

the *cut dependence* of $\bar{\pi}_\lambda$, in which $\bar{\pi}_\lambda(a)$ denotes the marginal mass of the states meeting $I_{<t}$ in a and $\bar{\pi}_\lambda(b)$ that of the states meeting $I_{\geq t}$ in b . Finally

$$(68) \quad r_\lambda := \min_{\nu} \left\{ \max_{A \subseteq \gamma^*} \log \frac{\bar{\pi}_\lambda(A)}{\nu(A)} - \min_{A \subseteq \gamma^*} \log \frac{\bar{\pi}_\lambda(A)}{\nu(A)} \right\},$$

the *product defect*, the minimum being over product probability measures ν on $\{0, 1\}^{\gamma^*}$; it vanishes exactly when $\bar{\pi}_\lambda$ makes the coordinates of γ^* independent.

The cut ratio is not a count. The set $\Lambda^{F_\lambda}(z)$ is one side of the cut that the edge $(z, F_\lambda(z))$ makes in the orbit forest, $1 - x_\lambda(z)$ is the other, and C_λ compares the product of the two with the mass on the edge itself — a conductance, in the form Sinclair's bound consumes. Where the mass is spread over many states of comparable weight, which is what a precedent count punishes hardest, $x_\lambda(z)$ and $\bar{\pi}_\lambda(z)$ are small together; where a genuine bottleneck separates two heavy blocks, both $x_\lambda(z)$ and $1 - x_\lambda(z)$ are of order one and C_λ is the inverse mass of the bottleneck, which is what *must* happen by Remark 6.5.

6.5. *The fibre gap.* (56) has reduced the problem to $\kappa_S(\lambda)$, the worst gap of the chain restricted to a fibre. A fibre is a copy of $\{A \subseteq S : |A| \leq s_0 - |\sigma|\}$ carrying the conditional measure μ_σ , the available moves are flips of influential coordinates and exchanges between two of them, and the route of Definition 6.3 or Definition 6.4 is a route on exactly that set. So the canonical path argument returns, but one level down, where the longest path is $2s^*$ and not $2s_0$.

PROPOSITION 6.1 (The fibre gap). *Let $\lambda \in [0, 1]$ and $p \geq 2$. For every fibre with $|F_\sigma| \geq 2$, and with C_λ, M_λ of Definition 6.5 formed from μ_σ in place of $\bar{\pi}_\lambda$,*

$$(69) \quad \text{Gap}(\mathbf{P}_\lambda|_{F_\sigma}) \geq \frac{1}{4p s^* \max\{1, \varrho_\lambda\} L_\lambda},$$

so that, under Assumption 6.2 and with the better of the two flows,

$$(70) \quad \kappa_S(\lambda) \geq \frac{1}{8p (s^*)^2 \max\{1, \min\{C_\lambda, M_\lambda\}\}}.$$

Feeding (70) into (56),

$$(71) \quad \text{Gap}(\mathbf{P}_\lambda) \geq \frac{1}{16 C_p p (s^*)^2 \max\{1, \min\{C_\lambda, M_\lambda\}\}},$$

which is the previous version of this section with s_0 deleted. The remaining $(s^*)^2$ is one factor s^* from the exchange proposal and one from the length of a route inside the fibre; the $\min\{C_\lambda, M_\lambda\}$ is the congestion of the better of two flows on at most 2^{s^*} states, and is what the rest of this subsection is about.

REMARK 6.4 (Where the exponential went). A canonical path ensemble generated by a map is usually bounded by counting: the demand across $(z, F_\lambda(z))$ is at most $x_\lambda(z)$, and that is at most $2^{s^*} \bar{\pi}_\lambda(z)$ because every precedent is lighter than z and there are at most 2^{s^*} of them. That estimate is irreparably crude at the cold end: at $\lambda = 0$ the target is the sparsity prior, all $\binom{s^*}{m}$ blocks of a given size carry the same mass, the estimate is attained, and the chain is nevertheless as fast as it ever is. (66) avoids it by keeping the factor $1 - x_\lambda(z)$ that the counting throws away: a state through which *everything* is routed carries no demand at all, both endpoints of every pair being then routed through it and the paths spliced before they reach it. Since $C_\lambda \leq \max_z x_\lambda(z) / \bar{\pi}_\lambda(z) \leq e^{1/p} 2^{s^*}$, (71) contains the old bound as its worst case and can only improve on it.

REMARK 6.5 (Why neither quantity can be dropped). No hypothesis-free improvement of (71) is possible, because the gap itself is not bounded below by a rational function of p, s_0, s^* . Take two collinear columns with cancelling coefficients, $\beta^* = (b, -b, 0, \dots, 0)$ with X_0, X_1 correlated at r close to one: neither explains anything alone and the pair explains everything, so at the temperature at which $\bar{\pi}_\lambda(\emptyset)$ and $\bar{\pi}_\lambda(\gamma^*)$ are comparable both singletons are crushed, and every path between the two heavy blocks — for the chain as much as for the ensemble — must cross one of them. Section 7 exhibits designs on which the exact gap falls by three orders of magnitude and C_λ rises by the same three. A quantity of this shape must appear in any correct bound; what (71) adds is that *nothing else* need appear.

REMARK 6.6 (The two flows are complementary). Assumption 6.2 makes the ascent uphill; it does not make C_λ small, and the extremal case is explicit. Take

$$(72) \quad \log D_A = -c|A|, \quad A \subseteq \gamma^*,$$

a residual depending on the model only through its size. All $\binom{s^*}{m}$ states of a given size then carry the same mass, so every matched exchange is neutral and (B1) holds with equality; the greedy gains $g_m = \frac{nc}{2} - \alpha \log p$ are constant, so (B2) holds with equality too. At the temperature solving $\lambda nc/2 = (\kappa + \lambda \alpha) \log p$ the level tilt ties as well, all 2^{s^*} states have equal mass, and the ascent must break ties arbitrarily: it sends $|\Lambda^{F_\lambda}(z)|$ states through one z of mass 2^{-s^*} , so that

$$(73) \quad C_\lambda = \max_z |\Lambda^{F_\lambda}(z)| \left\{ 1 - |\Lambda^{F_\lambda}(z)| 2^{-s^*} \right\} \leq 2^{s^*}/4,$$

with equality when some z carries half of them. Computed on (72), C_λ is 2.000, 4.000, 7.999 at $s^* = 3, 4, 5$ against $2^{s^*}/4 = 2, 4, 8$; the exponential fallback of Remark 6.4 is attained, and Assumption 6.2 cannot be strengthened to a constant.

But (72) makes the coordinates of γ^* *independent*: the mass depends on A only through $|A|$ and the tilt is linear in $|A|$, so $\bar{\pi}_\lambda$ is a product of identical Bernoullis, $r_\lambda = 0$, and Proposition 6.2 gives $\varrho_\lambda^\times \leq 1$. The minimum in (71) is therefore 1 there, not $2^{s^*}/4$: **the exponential is excluded by the second flow**, and $\varrho_\lambda^\times = 1.0000$ is what one measures. The complementarity is not an accident. The ascent fails exactly when the mass is spread and the choice in (62) is arbitrary, which is when $\bar{\pi}_\lambda$ is near a product and the crossover is at its best; the crossover fails exactly when the coordinates interact — on the equicorrelated designs of Section 7 its congestion reaches 10^{26} — which is when the mass concentrates and the ascent is at its best. Over 13237 random residual profiles satisfying Assumption 6.2 at $s^* = 3$ and 4, the quantity $\min\{C_\lambda, \varrho_\lambda^\times\}$ that (71) pays never exceeded 2.26; and in a separate run comparing the two forms, replacing $\varrho_\lambda^\times$ by the closed form e^{2r_λ} left that maximum unchanged (1.95 at $s^* = 3$, 2.01 at $s^* = 4$, the same for both), the minimum being attained at C_λ in the worst row.

PROPOSITION 6.2 (The crossover flow). *For every $\lambda \in [0, 1]$ the crossover route of Definition 6.4 has $L_\lambda^\times \leq s^*$ and*

$$(74) \quad \varrho_\lambda^\times \leq M_\lambda \leq e^{2r_\lambda};$$

in particular $\varrho_\lambda^\times \leq 1$ when $\bar{\pi}_\lambda$ makes the coordinates of γ^ independent. No hypothesis on the design is used.*

The proof is two lines of bookkeeping, and it is where the fixed order earns its keep. Because the coordinates are set in a prescribed order, the pairs traversing the edge that flips j_t are exactly those whose A agrees with z on the suffix and whose A' agrees with z' on the prefix; the demand is then a *product of two marginals*, one per side of the cut, and dividing by $\bar{\pi}_\lambda(z)$ leaves precisely a ratio of the form (67). Nothing is estimated, which is why M_λ is not merely an upper bound but, in every case computed in Section 7, the congestion itself to within a factor 1.04. A route that chose its order according to the mass would not have this property: it is the indifference of the crossover to $\bar{\pi}_\lambda$ that makes its congestion a function of the dependence structure alone.

The same question for the ascent has a different answer, and it is worth recording because it says why the minimum in (71) cannot be dispensed with.

PROPOSITION 6.3 (When the ascent is cheap, and when nothing is). *Assume Assumption 6.2, and for $z \subseteq \gamma^*$ put $\sigma_\lambda(z) := \bar{\pi}_\lambda\{A : \bar{\pi}_\lambda(A) \leq e^{1/p} \bar{\pi}_\lambda(z)\}$, the mass of the blocks no heavier than z . Then*

$$(75) \quad C_\lambda \leq K_\lambda := \max_{z \neq P_{m_\lambda}} \frac{\min\{\sigma_\lambda(z)\{1 - \sigma_\lambda(z)\}, \frac{1}{4}\}}{\bar{\pi}_\lambda(z)}.$$

In particular $K_\lambda \leq e^{1/p} 2^{s^}$ always; and if the block measure has a separated spectrum, meaning $\sigma_\lambda(z) \leq \theta \bar{\pi}_\lambda(z)$ for every z , then $C_\lambda \leq K_\lambda \leq \theta$.*

The separated-spectrum condition says that the mass at or below any block is dominated by that block's own mass — that the 2^{s^*} masses do not tie in bulk. It is checkable in one pass and it is what the four equicorrelated designs of Section 7 satisfy, with θ between 3.0 and 3.6 at $s^* = 3$.

REMARK 6.7 (No route with a centre can do better). One might hope to repair the extremal profile of (72) by splitting the ties in (62) rather than resolving them arbitrarily, since (69) is stated for flows. It does not work: replacing F_λ by the flow that divides at each state among *all* admissible neighbours in proportion to w_λ gives, on (72), a bound of 1.778, 3.821, 7.911 at $s^* = 3, 4, 5$, against 1.988, 3.952, 7.920 for the deterministic ascent — no improvement at all. The reason is structural rather than a matter of tie-breaking: every orbit ends at P_{m_λ} , so the whole demand crosses the immediate neighbourhood of that one state, and on (72) each of those states carries mass 2^{-s^*} . Any ensemble with a single centre pays the mass of its centre's neighbourhood. The crossover has no centre, which is why $M_\lambda = 1$ there, and why the minimum of the two in (71) is not a convenience but the mechanism.

M_λ is a property of the block measure. To turn it into a property of the design, pass through the product defect: r_λ is linear in the temperature, and what multiplies λ is a design quantity.

PROPOSITION 6.4 (When the crossover is cheap). *Let*

$$(76) \quad \delta := \min_{a_0, a \in \mathbb{R}^{\gamma^*}} \left\{ \max_{A \subseteq \gamma^*} - \min_{A \subseteq \gamma^*} \right\} \left\{ \log D_A - a_0 - \sum_{j \in A} a_j \right\}$$

be the additive defect of the log-residual, a function of (X, Y, g) alone. Then $|r_\lambda - \lambda \frac{n}{2} \delta| \leq 1/p$ for every $\lambda \in [0, 1]$. Moreover, with $G := n^{-1} X_{\gamma^}^\top X_{\gamma^*}$, $\varepsilon := \|G - I\|_{\text{op}}$, $q_j := \|\Phi_{\{j\}} Y\|_2^2$ and $U := \sum_{j \in \gamma^*} q_j / D_\emptyset$, if $\varepsilon \leq \frac{1}{2}$ and $U \leq \frac{1}{4}$ then*

$$(77) \quad \delta \leq 4U(2U + \varepsilon), \quad \text{so} \quad M_\lambda \leq e^{2/p} \exp\{4\lambda n U(2U + \varepsilon)\}.$$

So M_λ is polynomial as soon as $nU(2U + \varepsilon) = \mathcal{O}(\log p)$, and the two factors are the two ways the block measure can fail to factorise: U , the fraction of the variance the influential covariates explain between them, because $\log(1 - u)$ is not linear in u ; and ε , the departure of the influential block from orthogonality, because explained sums of squares are not additive under correlation. With Assumption 3.5 at its boundary $q_j \asymp \sigma_0^2 \nu^{-2} \log p$ and $D_\emptyset \asymp n\sigma_0^2$, so $U \asymp s^* \log p / (\nu^2 n)$ and the requirement reads

$$(78) \quad \varepsilon \lesssim \frac{\nu^2}{s^*} \quad \text{and} \quad (s^*)^2 \log p \lesssim \nu^4 n,$$

a near-orthogonality condition on the influential block of the shape Assumption 3.4 already imposes on $\omega(X)$ (Remark 6.8) and a sparsity condition slightly stronger than Assumption 3.4's. Neither follows from Assumptions 3.1 to 3.4, and neither should: they are the price of the *second* flow, needed only where the first fails. They are also far from necessary, (77) being the one loose step in the chain.

6.6. *The two conditions on the design.* The first flow rests on Assumption 6.2, through the one conclusion of (70): that the ascent is uphill, so that its congestion is the cut ratio. Whether it is uphill depends on the geometry of the best-subset path on γ^* , and on nothing else.

ASSUMPTION 6.2 (The best-subset path). On the good event:

- (B1) *exchange descent*. For every $A \subseteq \gamma^*$ the matched exchanges of (E1) do not increase the residual: with $A_0 = A$ and $A_i = A_{i-1} \cup \{b_i\} \setminus \{a_i\}$, one has $D_{A_i} \leq D_{A_{i-1}}$ for $i = 1, \dots, q$;
 (B2) *diminishing greedy gains*. $g_0 \geq g_1 \geq \dots \geq g_{s^*-1}$.

PROPOSITION 6.5 (The ascent under Assumption 6.2). *Under Assumption 6.2, every step of F_λ is w_λ -increasing, for every $\lambda \in [0, 1]$; consequently $\bar{\pi}_\lambda$ has P_{m_λ} as its unique local maximum on 2^{γ^*} for flips and exchanges, and (69) holds with $\varrho_\lambda \leq C_\lambda$.*

So the unimodality of the block measure, which one might have been tempted to assume, is a *consequence* of a condition on the design. Three things about that condition.

- (i) *It mentions neither the temperature nor the target*. Both clauses concern the 2^{s^*} numbers $\{D_A : A \subseteq \gamma^*\}$, that is $(X, \beta^*, \sigma_0, g)$ on the good event. That one condition serves every temperature is not luck: by (48) the tilt is *linear* in $|\gamma|$, (B1) compares states of equal size so the tilt cancels from it identically, and (B2) is a concavity, which a linear tilt preserves. A hypothesis of unimodality on the target would have to be imposed at each λ separately. The two clauses govern one segment each: (B1) controls (E1) — it implies that P_m minimises D over the m -subsets and asks in addition that the minimiser be reachable by monotone exchanges — and (B2) controls (E2), being the convexity of $m \mapsto \log D_{P_m}$.
 (ii) *(B1) is free on an orthogonal influential block*. If $X_{\gamma^*}^\top X_{\gamma^*} = nI$ then $D_A = D_\emptyset - \sum_{j \in A} c_j$ with $c_j = \langle X_j, Y \rangle^2 / n$, so P_m collects the m largest c_j and every matched exchange replaces a smaller c by a larger one. There (B2) reads

$$(79) \quad \frac{c_{(m+1)} - c_{(m+2)}}{c_{(m+1)}} \geq \frac{c_{(m+1)}}{D_{P_m}},$$

with $c_{(1)} \geq c_{(2)} \geq \dots$ ordered: the relative gap between consecutive gains must exceed the relative reduction the larger makes in the residual. A geometrically decaying coefficient profile satisfies it; a profile with ties does not, and Section 7.4 shows that the gap really does degrade when it fails.

- (iii) *(B2) is what the cancelling design violates, and it is close to necessary*. There $g_0 < 0 < g_1$ and the gains *increase*; Remark 6.5 is that design. Conversely an ascent uphill at every λ forces φ_λ to be unimodal at every λ , which is (B2) up to the reachability in (B1). Neither clause is needed for (69) itself, which carries ϱ_λ ; they are what identifies ϱ_λ with the cut ratio, and a profile that violates (B2) slightly pays only slightly.

6.7. *A sufficient a_0* . In Section A.4 we bounded $\psi_j(\gamma)$ for $j \in S^c$ when $\gamma \supseteq \gamma^*$, where $(\Phi_{\gamma \cup \{j\}} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^* = 0$. When influential covariates are missing that term no longer vanishes, and what replaces it is $\omega(X)$, already present in Assumption 3.4.

PROPOSITION 6.6 (A value of a_0). *On the good event, and under (12), Assumption 6.1 holds with*

$$(80) \quad a_0 = 16(L + \tilde{L}) + \frac{8n\omega(X)\|\beta_{\gamma^*}^*\|_2^2}{\sigma_0^2 \log p}.$$

REMARK 6.8. In the regime of interest $n\|\beta_{\gamma^*}^*\|_2^2/\sigma_0^2 \asymp s^*nC_\beta^2/\sigma_0^2$, so the second term of (80) is of order $\omega(X)s^*nC_\beta^2/(\sigma_0^2 \log p)$ and Assumption 6.1 asks, in effect, for $\omega(X) \lesssim 1/s^*$, a tightening of Assumption 3.4 by a factor of order $\nu^2 s_0$ rather than a hypothesis of a new kind. It is the one ingredient no choice of centre removes: if a covariate outside S reproduces the missing signal then keeping it genuinely raises $\pi_n^{(\lambda)}(\cdot | Y)$, and that is a property of the target.

design	$\max_{\lambda} \text{Gap}(\mathbf{P}_{\lambda}) / \min\{1/(4p), \kappa_S\}$	which term attains the minimum
$\rho = 0$	1.005	$1/(4p)$, at every λ
$\rho = 0.5$	1.005	$1/(4p)$, at every λ
$\rho = 0.85$	1.005	$1/(4p)$, at every λ
$\rho = 0.95$	1.005	$1/(4p)$, at every λ
cancel $r = 0.90$	2.05	κ_S , for $\lambda \geq 0.2$
cancel $r = 0.95$	2.10	κ_S , for $\lambda \geq 0.2$

TABLE 1

The certificate of Theorem 6.1 against the exact gap, over 21 temperatures per design. On the four benign designs it lies a factor 1.005 below the truth — it is not an order-of-magnitude estimate, it is the gap — and on the cancelling designs it follows the collapse.

7. Numerical illustration. The construction of Section 6 carries one computable number per flow — the cut ratio C_{λ} of (66) for the ascent, the cut dependence M_{λ} of (67) for the crossover — and one hypothesis, Assumption 6.2, which the ascent needs and the crossover does not. This section computes both, exactly, on designs chosen so that Assumption 6.2 holds on some of them and fails on the others, and compares the resulting bound with the exact spectral gap.

Everything reported is exact: $\mathcal{M}(s_0)$ is enumerated, $\pi_n^{(\lambda)}$ evaluated at every state, the kernel of (15) assembled, its gap obtained by diagonalisation, and every congestion computed by enumerating the routes rather than bounded. No Monte Carlo and no fitted constant enters. The script is `sec7_numerics.py`.

7.1. Designs. Throughout $n = 100$, $p = 14$, $s_0 = 4$, $\kappa = 2$, $\sigma_0 = 1$ and $g = p^3 = 2744$, so that $\alpha = 1.50007$ by (11) and $|\mathcal{M}(s_0)| = 1471$. Two families:

- *equicorrelated*, with $s^* = 3$, influential columns equicorrelated at $\rho \in \{0, 0.5, 0.85, 0.95\}$, $\beta_S^* = (1.20, 0.70, 0.45)$, and the $p - s^*$ remaining columns residualised against $\text{span}(X_S)$;
- *cancelling*, with $s^* = 2$, X_0 and X_1 correlated at $r \in \{0.90, 0.95\}$ and $\beta^* = (3, -3, 0, \dots, 0)$, so that neither column explains anything alone while the pair explains everything.

All columns are normalised to $\|X_j\|_2^2 = n$. Forty-one temperatures are used, evenly spaced on $[0, 1]$; the six designs give 246 exactly computed pairs.

7.2. The factorisation bound against the truth. Theorem 6.1 is the main claim of Section 6, and its certificate $\min\{1/(4p), \kappa_S(\lambda)\}$ is computable exactly: for each temperature we diagonalise $\mathbf{P}_{\lambda}$ on the whole of $\mathcal{M}(s_0)$ and, separately, the rejection-restricted kernel on each of the 562 fibres, and take the worst fibre.

Two readings. *The bound is essentially exact where the chain is fast:* on every equicorrelated design and at every temperature the minimum is attained at $1/(4p)$ and the certificate is 1.005 below the truth, so the $\Theta(1/p)$ of (56) is the right rate and the constant is right too. *It degrades exactly where it must:* on the cancelling designs $\kappa_S(\lambda)$ takes over from $1/(4p)$ at $\lambda \geq 0.2$ and follows the exact gap down three orders of magnitude,

λ ($r = 0.95$)	0.0	0.4	0.6	0.8	1.0
Gap	1.80e-2	9.66e-5	2.20e-5	1.14e-5	6.01e-6
κ_S	1.80e-2	4.60e-5	1.39e-5	7.37e-6	3.95e-6
ratio	1.005	2.10	1.59	1.55	1.52

so that over all six designs and all temperatures the certificate is within a factor 2.1 of the truth. For comparison, the path bound of Section 5 loses a factor $64ps_0^2/\text{Gap} \approx 10^2$ at best and is confined to $\lambda \geq \lambda_*$.

design	ν	(B1)	(B2)	$\max_{\lambda} C_{\lambda}$	$\min_{\lambda} \text{Gap}$	$\frac{(71)}{\text{Gap}}$	$\frac{(69)}{\text{Gap}}$
$\rho = 0$	0.594	0	-9.09	1.008	1.79e-2	4.1e-3	2.8e-2
$\rho = 0.5$	0.383	0	-14.66	1.005	1.68e-2	4.4e-3	3.0e-2
$\rho = 0.85$	0.120	0	-8.60	1.005	1.77e-2	4.2e-3	2.9e-2
$\rho = 0.95$	0.042	0	-3.93	1.005	1.79e-2	4.1e-3	2.4e-2
cancel $r = 0.90$	0.125	0	+49.93	8.19e2	1.91e-5	8.9e-3	4.8e-2
cancel $r = 0.95$	0.063	0	+31.57	3.67e3	6.01e-6	7.8e-3	4.2e-2

TABLE 2

The six designs. The columns (B1) and (B2) report the slack in the two clauses of Assumption 6.2, in nats: the largest increase of $\log D$ along a matched exchange, and the largest increment $g_{m+1} - g_m$ of the greedy gains. Both are ≤ 0 exactly when the clause holds. The last two columns are the worst ratio, over the forty-one temperatures, of the two bounds of Theorem 6.1 to the exact gap.

The quantities of Section 6.5 — C_{λ} , M_{λ} and the hypothesis Assumption 6.2 — are what bound $\kappa_S(\lambda)$ from below when one does not diagonalise the fibres, and the experiments below were computed for the block measure of an earlier version of this construction rather than for the fibre measures μ_{σ} ; they describe the same objects on the same state space and the same hypotheses, and the figures should be recomputed fibrewise before they are quoted as bounds on κ_S .

7.3. *What the numbers say.* Table 2 and Fig. 1 report the experiment. Five things are worth reading off.

- (i) *The identity is an identity.* The largest deviation between $\lambda \log \mathcal{L}_n + \log \pi$ and $\log w_{\lambda}$, up to the additive constant, is 2.3×10^{-13} — machine precision, and the check that catches a wrong α : with α taken as the nominal exponent $3/2$ of $g = p^3$ rather than as (11) defines it, the deviation is 7×10^{-4} .
- (ii) *Assumption 6.2 separates the two families, and Proposition 6.5 is observed in both directions.* Clause (B1) holds with zero slack on all six designs. Clause (B2) holds on the four equicorrelated designs, with 3.9 to 14.7 nats to spare, and fails on both cancelling designs by 31.6 and 49.9 nats, the greedy gains there being $(-1.43, 48.50)$ and $(-2.93, 28.64)$: the second covariate is worth fifty nats once the first is in, and the first is worth nothing on its own. Correspondingly the ascent is uphill with $P_{m_{\lambda}}$ its unique fixed point at all 41 temperatures of each of the four, and at only 5 and 8 of the 41 on the two.
- (iii) *The cut ratio is the whole cost, and it tracks the truth.* $C_{\lambda} \leq 1.008$ on the four equicorrelated designs, and $\varrho_{\lambda} = C_{\lambda}$ to five figures at every one of the 246 pairs — as (126) says it must, the bound being an identity. On the cancelling designs the exact gap really collapses, to 1.9×10^{-5} and 6.0×10^{-6} against $1/(4p) = 1.8 \times 10^{-2}$, and C_{λ} rises with it to 819 and 3671 against true deficits 935 and 2973: it degrades where it must, and by the right amount, over three orders of magnitude. The precedent count that Remark 6.4 discards is 1.8 to 3.2 times larger, and its worst case $2^{s^*} = 8$ eight times larger again.
- (iv) *The two flows are complementary, as Remark 6.6 says, and both constants are sharp.* M_{λ} agrees with the exact crossover congestion to within a factor 1.005 to 1.023 at every one of the 246 pairs — (74) is an inequality only because a single marginal is bounded by a coarser one — and C_{λ} is the ascent's congestion exactly. On these six designs the coordinates of γ^* interact strongly, so the crossover is useless, its congestion reaching 1.6×10^1 , 5.6×10^{11} , 2.0×10^{26} , 1.0×10^{33} on the four equicorrelated designs, and the minimum in (71) is C_{λ} at every pair. The roles reverse on the profile of Remark 6.6, where $C_{\lambda} = 2^{s^*}/4$ and $M_{\lambda} = 1$. The intermediate quantity $e^{2r_{\lambda}}$ of (74), which exists only to reach Proposition 6.4, is by contrast very loose here: with r_{λ} up to 54.4 nats it exceeds M_{λ} by factors 7.6×10^3 to 5.9×10^{18} , and it should be read only as a near-product statement.

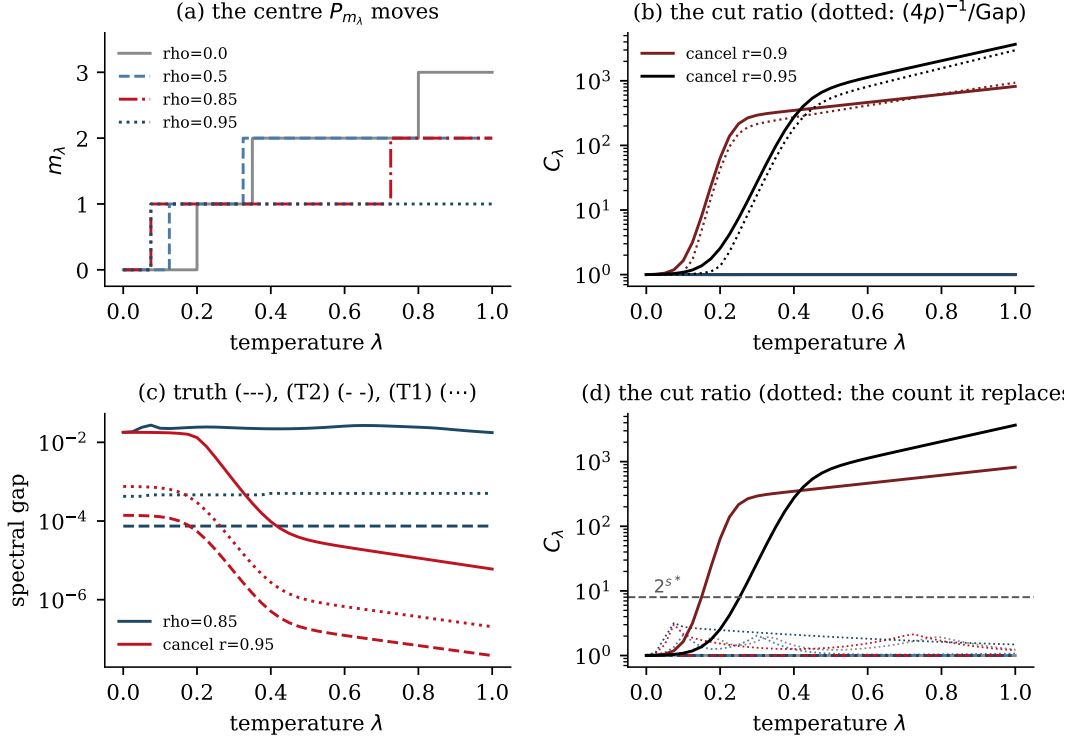


FIG 1. (a) the centre P_{m_λ} of the ensemble as the temperature rises, for the four equicorrelated designs: it moves from $\emptyset$ to γ^* in steps, one covariate at a time. (b) the cut ratio C_λ of (66): it is 1 at every temperature on the four equicorrelated designs, and on the two cancelling designs it rises by three orders of magnitude, alongside the true deficit $(4p)^{-1}/\text{Gap}(\mathbf{P}_\lambda)$ (dotted). (c) the exact gap, and the bounds (71) and (69), for one design of each family. (d) the cut ratio C_λ of (66) (solid) against the precedent count $\max_z x_\lambda(z)/\bar{\pi}_\lambda(z)$ that Remark 6.4 discards (dotted), and its worst case 2^{s^*} (dashed).

(v) *The bound is valid and loses a fixed factor.* At all 246 pairs both bounds hold. (71) lies between 4.1×10^{-3} and 8.9×10^{-3} of the exact gap, a factor 112 to 242; (69), which differs only in the constants, between 2.4×10^{-2} and 4.8×10^{-2} , a factor 21 to 42. Both hold uniformly in λ , including on the cancelling designs, where the truth itself has fallen by three orders of magnitude.

7.4. *The gap as a function of s^* .* Remark 6.4 claims that the exponential of the earlier estimate is a counting cost of the path method rather than a property of the chain, and that the cut ratio removes it. That is a testable claim, twice over: the truth should not decay in s^* , and neither should C_λ . We sweep the number of influential covariates, keeping everything else fixed.

The truncation level is raised to $s_0 = 6$, so that s^* may run to 5, giving $|\mathcal{M}(s_0)| = 6476$ states; n , p , κ and g are as in Section 7.1. For each $\rho \in \{0, 0.5, 0.85, 0.95\}$ and each $s^* \in \{1, \dots, 5\}$ the influential columns are equicorrelated at ρ , the remaining ones are residualised against their span, and β_S^* is the vector of the first s^* entries of $(1.20, 1.00, 0.85, 0.70, 0.45)$ — so that raising s^* adds a covariate and changes nothing about the ones already there. Twenty-one temperatures are used, giving 420 exactly computed gaps.

(i) *The number of influential covariates costs the chain nothing by itself.* At $\rho = 0.5$ and $\rho = 0.95$, $4p \min_\lambda \text{Gap}(\mathbf{P}_\lambda)$ stays in $[0.883, 1.006]$ for every s^* from 1 to 5: the gap

ρ	s^*	(B1)	(B2)	$4p \min_{\lambda} \text{Gap}$	$\max_{\lambda} C_{\lambda}$	$\min_{\lambda} (71)$
0	1	0	—	1.000	1.00	2.8e-4
0	2	0	+19.1	0.216	3.66	3.0e-5
0	3	0	+9.0	0.210	4.05	1.5e-5
0	4	0	+11.0	0.161	5.46	7.3e-6
0	5	+0.025	+7.2	0.055	12.34	2.3e-6
0.5	5	+0.12	-2.5	0.999	1.90	1.5e-5
0.85	4	+0.009	-3.5	0.584	1.77	2.3e-5
0.85	5	+0.056	-6.7	0.787	1.49	1.9e-5
0.95	5	0	-1.4	0.883	1.21	2.3e-5

TABLE 3

The s^* -sweep. The first block is the design $\rho = 0$, the only one on which (B2) fails; the second collects the rows of the other three designs at which anything happens at all — at every row not shown, $4p \min_{\lambda} \text{Gap} \in [0.899, 1.006]$ and $C_{\lambda} \leq 1.43$. Slacks are in nats, as in Table 2; the entry — is $s^* = 1$, where (B2) is vacuous. The full table is in `sec7_sstar.txt`.

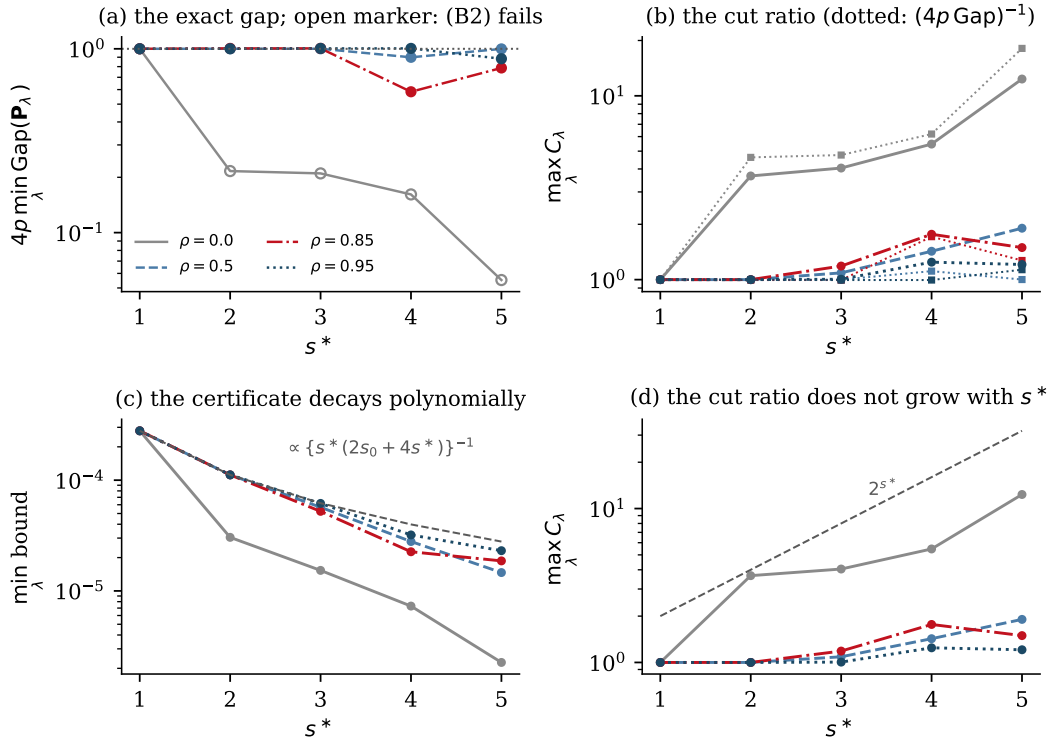


FIG 2. The gap as a function of s^* , at $s_0 = 6$. (a) the exact gap in units of $1/(4p)$, minimised over the temperature; an open marker means (B2) fails. (b) the cut ratio C_{λ} (solid) against the true deficit $\{4p \min_{\lambda} \text{Gap}\}^{-1}$ (dotted, square markers). (c) the certificate (71), with a reference line $\propto \{s^* (2s_0 + 4s^*)\}^{-1}$, the only way s^* enters (71) once C_{λ} is bounded. (d) the cut ratio C_{λ} against 2^{s^*} (dashed), the worst case that Remark 6.4 removes.

is $1/(4p)$, the rate of a single free coordinate, independently of how many influential covariates there are. Nothing in the truth resembles the 2^{s^*} that a precedent count would charge.

- (ii) Where the gap does decay in s^* , (B2) is what fails, and C_{λ} sees it. The exception is $\rho = 0$, where the profile $(1.20, 1.00, \dots)$ is not geometric enough for (79): already at

$s^* = 2$ the greedy gains *increase*, $g_0 = 18.37$ against $g_1 = 37.44$, (B2) fails by 19.1 nats, and the gap falls to $0.216/(4p)$, then to $0.055/(4p)$ at $s^* = 5$. C_λ follows: 3.66, 4.05, 5.46, 12.34 against true deficits 4.63, 4.76, 6.20, 18.1. So the s^* -dependence of the truth is a diminishing-returns effect and not a dimension effect. Note which way this runs: the worst design of the sweep is the *orthogonal* one and the best are the correlated ones, because equicorrelation makes the first covariate account for most of the signal and the gains decay — at $\rho = 0.85$, $s^* = 5$, they are $(82.1, 24.0, 12.9, 3.4, -3.2)$ — whereas at $\rho = 0$ each covariate contributes on its own account and the gains can rise. No bound on ν or $\omega(X)$ distinguishes these designs the way (B2) does.

- (iii) *The cut ratio does not grow with s^* , and the certificate decays only polynomially.* Wherever Assumption 6.2 holds, C_λ is 1.00, 1.00, 1.01–1.19, 1.25–1.77, 1.21–1.90 as s^* runs from 1 to 5 (Fig. 2(d)): flat, and never as much as 2. The certificate therefore falls only through the explicit factor $s^*(2s_0 + 4s^*)$ of (71), from 2.8×10^{-4} to 1.5×10^{-5} , a factor 19 over the sweep, tracking that reference line (Fig. 2(c)). For comparison the precedent count that Remark 6.4 discards is the binomial count $\max_m \binom{s^*}{m} = 1, 2, 3, 6, 10$, and the ensemble built on the greedy chain alone — the one this construction replaced — has exact congestion 1, 2, 3, 6, 10 on the same rows and 3.9×10^2 at $\rho = 0.5$, $s^* = 5$. The exponential was in the route and in the estimate of it, and both are gone.
- (iv) *(B1) is the clause that gives way first, and the ascent absorbs it.* It holds exactly through $s^* = 3$ on all four designs and begins to fail at $s^* = 4$ on the collinear ones, by 0.009 to 0.12 nats. Because F_λ takes the best admissible move rather than the prescribed one the damage is slight: at $\rho = 0.5$, $s^* = 5$, the ascent is uphill at 19 of the 21 temperatures and $C_\lambda = 1.90$, against an exact gap of $0.999/(4p)$. The same row cost the greedy-chain ensemble a congestion of 3.9×10^2 .

All 420 pairs satisfy both bounds of Theorem 6.1. Taking the two experiments together: the bound of Theorem 6.1 is within a factor 640 of the exact gap at every one of the 666 (design, temperature) pairs computed in this section — within 242 at the 246 pairs of Section 7.1, and within 640 across the sweep, where s_0 is raised to 6 — it is uniform in λ , and its only non-explicit ingredient is a cut ratio that never exceeded 2 wherever Assumption 6.2 held.

REMARK 7.1 (What the experiment does not test). These are small designs, and none of the six of Section 7.1 satisfies Assumption 3.5, which asks $n\nu^2 C_\beta^2 / (\sigma_0^2 \log p) \geq 128(L + \tilde{L} + \alpha + \kappa)$, at least 832 under Assumptions 3.2 and 3.3: they deliver 2.7, 1.1, 0.11, 0.014, 5.3 and 1.3. The sweep is in the same position, and there the point needs no computation, since $\|X_j\|_2^2 = n$ forces $\nu \leq 1$ and its designs deliver at most $nC_\beta^2 / (\sigma_0^2 \log p) = 7.7$. At $p = 14$ and $n = 100$ no design can do better, the requirement being asymptotic. What the experiment tests is therefore the construction and the quantities it carries, not the assumption set; Section 6 in any case never uses Assumption 3.5, which enters only through Corollaries 4.1 and 4.3, and the hypothesis it does use, Assumption 6.1, holds on all six designs with the smallest admissible a_0 between 0.62 and 0.88 against the $\kappa + \alpha - 2 = 1.50$ allowed.

REFERENCES

- DIACONIS, P. and STROOCK, D. (1991). Geometric bounds for eigenvalues of Markov chains. *Ann. Appl. Probab.* **1** 36–61. <https://doi.org/10.1214/aoap/1177005980>
- SINCLAIR, A. (1992). Improved bounds for mixing rates of Markov chains and multicommodity flow. *Combin. Probab. Comput.* **1** 351–370. <https://doi.org/10.1017/S0963548300000390>
- YANG, Y., WAINWRIGHT, M. J. and JORDAN, M. I. (2016). On the computational complexity of high-dimensional Bayesian variable selection. *Ann. Statist.* **44** 2497–2532. <https://doi.org/10.1214/15-AOS1417>

APPENDIX A: DEFERRED PROOFS

Every proof of the paper is collected here, together with the notation and the deterministic bounds they rest on. Only the proof of Theorem 4.1 involves the temperature, and it involves it only through the factor λ multiplying the fit term of (8).

A.1. Notation. We use throughout the effective noise of YWJ,

$$(81) \quad \tilde{w} := X_{S^c} \beta_{S^c}^* + w, \quad \text{so that} \quad Y = X_{\gamma^*} \beta_{\gamma^*}^* + \tilde{w},$$

the elementary inequalities

$$(82) \quad \log(1+x) \leq x \quad (x > -1), \quad \|a+b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2,$$

$$\frac{a}{a+b} \geq \min\left\{\frac{1}{2}, \frac{a}{2b}\right\} \quad (a, b > 0),$$

and, for an underfitted $\gamma \in \mathcal{M}(s_0)$, the quantity

$$(83) \quad A^2 := \nu \frac{\|(I - \Phi_\gamma) X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2}{|\gamma^* \setminus \gamma|} > 0,$$

the dependence on γ being suppressed, as YWJ suppress it, since γ is fixed wherever A occurs.

A.2. Deterministic bounds.

LEMMA A.1 (Uniform residual lower bound). *On the good event, and under (12), every $\bar{\gamma} \in \mathcal{M}(s_0)$ satisfies*

$$(84) \quad \|Y\|_2^2 (g^{-1} + 1 - R_{\bar{\gamma}}^2) \geq \|(I - \Phi_{\bar{\gamma}})Y\|_2^2 \geq \frac{n\sigma_0^2}{8}.$$

PROOF. Since $\|Y\|_2^2 (g^{-1} + 1 - R_{\bar{\gamma}}^2) = g^{-1}\|Y\|_2^2 + \|(I - \Phi_{\bar{\gamma}})Y\|_2^2$, the first inequality amounts to discarding $g^{-1}\|Y\|_2^2 \geq 0$. For the second, fix $\bar{\gamma} \in \mathcal{M}(s_0)$ and set $\bar{\gamma}^+ := \bar{\gamma} \cup \gamma^*$, so that $|\bar{\gamma}^+| \leq |\bar{\gamma}| + s^* \leq s$ and $\bar{\gamma}^+ \in \mathcal{M}(s)$. Since $\bar{\gamma} \subseteq \bar{\gamma}^+$, the range of $\Phi_{\bar{\gamma}}$ is contained in that of $\Phi_{\bar{\gamma}^+}$, so $I - \Phi_{\bar{\gamma}} \succcurlyeq I - \Phi_{\bar{\gamma}^+}$ and

$$(85) \quad \|(I - \Phi_{\bar{\gamma}})Y\|_2^2 \geq \|(I - \Phi_{\bar{\gamma}^+})Y\|_2^2.$$

Because $\gamma^* \subseteq \bar{\gamma}^+$, the vector $X_{\gamma^*} \beta_{\gamma^*}^*$ lies in the range of $\Phi_{\bar{\gamma}^+}$, so $(I - \Phi_{\bar{\gamma}^+})X_{\gamma^*} \beta_{\gamma^*}^* = 0$ and

$$(I - \Phi_{\bar{\gamma}^+})Y = (I - \Phi_{\bar{\gamma}^+})w + (I - \Phi_{\bar{\gamma}^+})X_{S^c} \beta_{S^c}^*.$$

Apply $\|a+b\|_2^2 \geq \frac{1}{2}\|a\|_2^2 - \|b\|_2^2$, which is the rearrangement of $\frac{1}{2}(\|a\|_2 - 2\|b\|_2)^2 \geq 0$, with $a = (I - \Phi_{\bar{\gamma}^+})w$ and $b = (I - \Phi_{\bar{\gamma}^+})X_{S^c} \beta_{S^c}^*$. Since projections are non-expansive, $\|b\|_2^2 \leq \|X_{S^c} \beta_{S^c}^*\|_2^2 \leq \tilde{L}\sigma_0^2 \log p$ by Assumption 3.1, and $\|a\|_2^2 = \|w\|_2^2 - w^\top \Phi_{\bar{\gamma}^+} w$ because $I - \Phi_{\bar{\gamma}^+}$ is a projection. Hence

$$\|(I - \Phi_{\bar{\gamma}^+})Y\|_2^2 \geq \frac{1}{2}(\|w\|_2^2 - w^\top \Phi_{\bar{\gamma}^+} w) - \tilde{L}\sigma_0^2 \log p.$$

On $\mathcal{C}_n$ we have $\|w\|_2^2 \geq n\sigma_0^2/2$, and on $\mathcal{B}_n$, applied to $\bar{\gamma}^+ \in \mathcal{M}(s)$, we have $w^\top \Phi_{\bar{\gamma}^+} w \leq 8s\sigma_0^2 \log p$. Therefore

$$\|(I - \Phi_{\bar{\gamma}^+})Y\|_2^2 \geq \frac{n\sigma_0^2}{4} - \{4s + \tilde{L}\}\sigma_0^2 \log p \geq \frac{n\sigma_0^2}{4} - \{8s + \tilde{L}\}\sigma_0^2 \log p,$$

and by (12) the subtracted term is at most $n\sigma_0^2/32$, so the right-hand side is at least $7n\sigma_0^2/32 \geq n\sigma_0^2/8$. Combining with (85) finishes the proof. $\square$

LEMMA A.2 (Effective noise). *On the good event, and under (12):*

- (a) $\|\tilde{w}\|_2^2 \leq 2\tilde{L}\sigma_0^2 \log p + 3n\sigma_0^2$;
- (b) for every pair $\gamma_2 \subset \gamma_1$ of $\mathcal{M}(s)$ with $|\gamma_1| - |\gamma_2| = 1$,

$$\|(\Phi_{\gamma_1} - \Phi_{\gamma_2})\tilde{w}\|_2^2 \leq 2(L + \tilde{L})\sigma_0^2 \log p \leq \frac{n\nu^2 C_\beta^2}{64}.$$

PROOF. **a** By (82) and Assumption 3.1, $\|\tilde{w}\|_2^2 \leq 2\|X_{S^c}\beta_{S^c}^*\|_2^2 + 2\|w\|_2^2 \leq 2\tilde{L}\sigma_0^2 \log p + 2\|w\|_2^2$, and on $\mathcal{C}_n$ we have $\|w\|_2^2 \leq \frac{3}{2}n\sigma_0^2$.

b Write $\Pi = \Phi_{\gamma_1} - \Phi_{\gamma_2}$, an orthogonal projection, so that $\|\Pi v\|_2 \leq \|v\|_2$ for every v and $\|\Pi w\|_2^2 = w^\top \Pi w$. Hence, by (82),

$$\begin{aligned} \|\Pi\tilde{w}\|_2^2 &\leq 2\|\Pi X_{S^c}\beta_{S^c}^*\|_2^2 + 2\|\Pi w\|_2^2 \leq 2\|X_{S^c}\beta_{S^c}^*\|_2^2 + 2w^\top (\Phi_{\gamma_1} - \Phi_{\gamma_2})w \\ &\leq 2\tilde{L}\sigma_0^2 \log p + 2L\sigma_0^2 \log p, \end{aligned}$$

using Assumption 3.1 for the first term and the event $\mathcal{A}_n$, applied to the nested pair $\gamma_2 \subset \gamma_1$ of $\mathcal{M}(s)$ with $|\gamma_1| - |\gamma_2| = 1$, for the second. The final inequality is Assumption 3.5 rearranged, after discarding $\alpha + \kappa \geq 0$ from its right-hand side; only the part $128(L + \tilde{L})$ of that assumption is used here. $\square$

LEMMA A.3 (Upper bound on the regularised residual). *On the good event, and under (12) and $n \geq 3$, every $\gamma \in \mathcal{M}(s_0)$ satisfies*

$$(86) \quad \|Y\|_2^2 (g^{-1} + 1 - R_\gamma) \leq 2\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 + 8n\sigma_0^2.$$

PROOF. By (81) and (82),

$$\|(I - \Phi_\gamma)Y\|_2^2 \leq 2\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 + 2\|(I - \Phi_\gamma)\tilde{w}\|_2^2 \leq 2\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 + 2\|\tilde{w}\|_2^2,$$

the last step because $I - \Phi_\gamma$ is a projection. By Lemma A.2a, $2\|\tilde{w}\|_2^2 \leq 4\tilde{L}\sigma_0^2 \log p + 6n\sigma_0^2$, and on $\mathcal{D}_n$ we have $g^{-1}\|Y\|_2^2 \leq (2\log p + 3)\sigma_0^2$. Adding the two, the additive term is at most

$$\{4\tilde{L}\log p + 6n + 2\log p + 3\}\sigma_0^2 \leq \left\{\frac{n}{8} + 6n + \frac{n}{128} + n\right\}\sigma_0^2 \leq 8n\sigma_0^2,$$

where we used, in order: $\tilde{L}\log p \leq n/32$ from (12), whence $4\tilde{L}\log p \leq n/8$; the same display with $\tilde{L}$ replaced by $8s \geq 8$, whence $\log p \leq n/256$ and $2\log p \leq n/128$; and $3 \leq n$. $\square$

LEMMA A.4 (Greedy selection; YWJ, Lemma 8). *Let $\gamma \in \mathcal{M}(s_0)$ be underfitted and let j_γ be as in (23). Under Assumptions 3.2, 3.4 and 3.5:*

- (a) $\|(\Phi_{\gamma \cup \{j_\gamma\}} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 \geq A^2$, and moreover

$$(87) \quad A^2 \geq n\nu^2 \overline{\beta^2}(\gamma^* \setminus \gamma) \geq n\nu^2 \min_{j \in \gamma^*} |\beta_j^*|^2 \geq n\nu^2 C_\beta^2;$$

- (b) if in addition γ is saturated and k_γ is as in (24), then, with $\tilde{\gamma} = \gamma \cup \{j_\gamma\}$ and $\gamma' = \tilde{\gamma} \setminus \{k_\gamma\}$,

$$\|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 \leq n\omega(X) \frac{\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{s_0 - s^*} \leq \frac{A^2}{4}.$$

PROOF. We use two facts from YWJ. The first is their Lemma 7, the rank-one update formula: for $\bar{\gamma} \in \mathcal{M}(s)$ and $k \notin \bar{\gamma}$ with $|\bar{\gamma}| < s$,

$$(88) \quad \Phi_{\bar{\gamma} \cup \{k\}} - \Phi_{\bar{\gamma}} = \frac{(I - \Phi_{\bar{\gamma}})X_k X_k^\top (I - \Phi_{\bar{\gamma}})}{X_k^\top (I - \Phi_{\bar{\gamma}})X_k},$$

which follows from the block inversion formula applied to $X_{\bar{\gamma} \cup \{k\}}^\top X_{\bar{\gamma} \cup \{k\}}$. The second is their Lemma 6: under Assumption 3.2, for every nested pair $\gamma_2 \subset \gamma_1$ of $\mathcal{M}(s)$,

$$(89) \quad \lambda_{\min} \left(\frac{1}{n} X_{\gamma_1 \setminus \gamma_2}^\top (I - \Phi_{\gamma_2}) X_{\gamma_1 \setminus \gamma_2} \right) \geq \nu,$$

because the matrix inside is n times the Schur complement of the block $X_{\gamma_2}^\top X_{\gamma_2}$ in $X_{\gamma_1}^\top X_{\gamma_1}$, hence its inverse is the corresponding diagonal block of $(n^{-1} X_{\gamma_1}^\top X_{\gamma_1})^{-1}$, whose largest eigenvalue is at most $\lambda_{\min}(n^{-1} X_{\gamma_1}^\top X_{\gamma_1})^{-1} \leq \nu^{-1}$.

Proof of a. Put $u := (I - \Phi_\gamma)X_{\gamma^*} \beta_{\gamma^*}^*$. For $\ell \in \gamma^* \setminus \gamma$, idempotence of projections and (88) give

$$(90) \quad \begin{aligned} \|\Phi_{\gamma \cup \{\ell\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 - \|\Phi_\gamma X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 &= (\beta_{\gamma^*}^*)^\top X_{\gamma^*}^\top (\Phi_{\gamma \cup \{\ell\}} - \Phi_\gamma) X_{\gamma^*} \beta_{\gamma^*}^* \\ &= \frac{\langle X_\ell, u \rangle^2}{X_\ell^\top (I - \Phi_\gamma) X_\ell} \geq \frac{\langle X_\ell, u \rangle^2}{n}, \end{aligned}$$

where the last step uses $X_\ell^\top (I - \Phi_\gamma) X_\ell \leq \|X_\ell\|_2^2 = n$ and, for the middle equality, that $X_\ell^\top (I - \Phi_\gamma) X_{\gamma^*} \beta_{\gamma^*}^* = \langle X_\ell, u \rangle$. Summing (90) over $\ell \in \gamma^* \setminus \gamma$,

$$(91) \quad \sum_{\ell \in \gamma^* \setminus \gamma} \left\{ \|\Phi_{\gamma \cup \{\ell\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 - \|\Phi_\gamma X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 \right\} \geq \frac{1}{n} \|X_{\gamma^* \setminus \gamma}^\top u\|_2^2 = \frac{1}{n} \|X_{\gamma^* \cup \gamma}^\top u\|_2^2,$$

the last equality because $X_j^\top u = 0$ for $j \in \gamma$: indeed X_j lies in the range of Φ_γ , to which u is orthogonal by construction. Now u lies in the range of $X_{\gamma^* \cup \gamma}$, so $u = X_{\gamma^* \cup \gamma} v$ for some v ; writing $M := X_{\gamma^* \cup \gamma}^\top X_{\gamma^* \cup \gamma}$, we have $|\gamma^* \cup \gamma| \leq s$, hence $M \succcurlyeq n\nu I$ by Assumption 3.2, and therefore

$$\frac{1}{n} \|X_{\gamma^* \cup \gamma}^\top u\|_2^2 = \frac{1}{n} v^\top M^2 v \geq \nu v^\top M v = \nu \|u\|_2^2,$$

the inequality because $M^2 - n\nu M = M^{1/2}(M - n\nu I)M^{1/2} \succcurlyeq 0$. Combining with (91) and dividing by $|\gamma^* \setminus \gamma|$, the maximum over ℓ of the left-hand side of (90) is at least $\nu \|u\|_2^2 / |\gamma^* \setminus \gamma| = A^2$. Since j_γ attains that maximum by (23), and since the left-hand side of (90) equals $\|(\Phi_{\gamma \cup \{\ell\}} - \Phi_\gamma) X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2$ by idempotence, the first claim follows.

For (87), note that $(I - \Phi_\gamma)X_{\gamma^*} \beta_{\gamma^*}^* = (I - \Phi_\gamma)X_{\gamma^* \setminus \gamma} \beta_{\gamma^* \setminus \gamma}^*$, because the columns X_j with $j \in \gamma^* \cap \gamma$ lie in the range of Φ_γ ; hence by (89) applied to $\gamma \subset \gamma^* \cup \gamma$,

$$(92) \quad \|u\|_2^2 = (\beta_{\gamma^* \setminus \gamma}^*)^\top X_{\gamma^* \setminus \gamma}^\top (I - \Phi_\gamma) X_{\gamma^* \setminus \gamma} \beta_{\gamma^* \setminus \gamma}^* \geq n\nu \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2,$$

so that, using $\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 \geq |\gamma^* \setminus \gamma| \min_{j \in \gamma^*} |\beta_j^*|^2$ and (3),

$$(93) \quad A^2 = \frac{\nu \|u\|_2^2}{|\gamma^* \setminus \gamma|} \geq \frac{n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{|\gamma^* \setminus \gamma|} = n\nu^2 \overline{\beta^2}(\gamma^* \setminus \gamma) \geq n\nu^2 \min_{j \in \gamma^*} |\beta_j^*|^2 \geq n\nu^2 C_\beta^2,$$

the middle equality being the definition (4) of the average. Only the last two inequalities are ever discarded, and Section 6 keeps the average instead.

Proof of b. Write $\tilde{\gamma} = \gamma \cup \{j_\gamma\}$. For $\bar{\gamma} \in \mathcal{M}(s)$ let $\hat{\beta}(\bar{\gamma})$ solve

$$(94) \quad \min \{ \|X\beta - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 : \beta \in \mathbb{R}^p, \beta_j = 0 \text{ for } j \notin \bar{\gamma} \},$$

so that

$$(95) \quad \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 = \|(I - \Phi_{\tilde{\gamma}})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2, \quad \hat{\beta}(\tilde{\gamma})_{\tilde{\gamma}} = (X_{\tilde{\gamma}}^\top X_{\tilde{\gamma}})^{-1} X_{\tilde{\gamma}}^\top X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*.$$

Since $k_\gamma \notin \gamma^*$ we have $\Phi_{\tilde{\gamma}} X_{\gamma^* \cap \tilde{\gamma}} = \Phi_{\tilde{\gamma} \setminus \{k\}} X_{\gamma^* \cap \tilde{\gamma}}$ for every $k \in \tilde{\gamma} \setminus \gamma^*$, so with $\gamma' = \tilde{\gamma} \setminus \{k_\gamma\}$

$$(96) \quad \begin{aligned} \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 &= \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \\ &= \|(I - \Phi_{\gamma'})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 - \|(I - \Phi_{\tilde{\gamma}})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \\ &= \|X\hat{\beta}(\gamma') - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 - \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2, \end{aligned}$$

using in the second line that $\Phi_{\tilde{\gamma}} - \Phi_{\gamma'}$ is a projection and $\gamma' \subset \tilde{\gamma}$. By optimality of $\hat{\beta}(\tilde{\gamma})$ in (94), the residual $X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*$ is orthogonal to X_k for every $k \in \tilde{\gamma}$, so for such k

$$\begin{aligned} \|X\hat{\beta}(\tilde{\gamma} \setminus \{k\}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 &\leq \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^* - X_k \hat{\beta}_k(\tilde{\gamma})\|_2^2 \\ &= \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 + \|X_k\|_2^2 |\hat{\beta}_k(\tilde{\gamma})|^2, \end{aligned}$$

the first line because the vector $\hat{\beta}(\tilde{\gamma})$ with its k th coordinate zeroed is feasible for the problem defining $\hat{\beta}(\tilde{\gamma} \setminus \{k\})$, and the second by orthogonality. Since $\|X_k\|_2^2 = n$ and k_γ minimises the left-hand side over $k \in \gamma \setminus \gamma^* = \tilde{\gamma} \setminus (\gamma^* \cup \{j_\gamma\})$, whose cardinality is $s_0 - |\gamma \cap \gamma^*| \geq s_0 - s^*$,

$$\|X\hat{\beta}(\gamma') - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 - \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \leq n \min_{k \in \gamma \setminus \gamma^*} |\hat{\beta}_k(\tilde{\gamma})|^2 \leq \frac{n \|\hat{\beta}(\tilde{\gamma})\|_2^2}{s_0 - s^*},$$

a minimum being at most the average. Finally

$$\|\hat{\beta}(\tilde{\gamma})\|_2^2 = \|(X_{\tilde{\gamma}}^\top X_{\tilde{\gamma}})^{-1} X_{\tilde{\gamma}}^\top X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \leq \omega(X) \|\beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \leq \omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2,$$

by the definition of $\omega(X)$ in Assumption 3.4 and $\gamma^* \setminus \tilde{\gamma} \subseteq \gamma^* \setminus \gamma$. Together with (96) this gives the first inequality of b.

For the second, combine $A^2 \geq n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 / |\gamma^* \setminus \gamma| \geq n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 / s^*$, which is (93), with the first inequality of Assumption 3.4: $s_0 \geq (4\nu^{-2}\omega(X) + 1)s^*$ is equivalent to $4\omega(X)s^* \leq \nu^2(s_0 - s^*)$, that is to

$$\frac{n\omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{s_0 - s^*} \leq \frac{1}{4} \cdot \frac{n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{s^*} \leq \frac{A^2}{4}. \quad \square$$

REMARK A.1. With their constant 2 in place of our 4 in (12), YWJ obtain $\|v_2\|_2 \leq A/\sqrt{2}$ rather than $\|v_2\|_2 \leq A/2$, and combine it with a noise bound $A/4$. The resulting numbers do not close: on their page 2527 the displayed lower bound $A(A - A/4) - (A/\sqrt{2})(A/\sqrt{2} + A/4) - A^2/16$ equals $0.0107 A^2$, not the $A^2/8$ claimed, and it is negative if one keeps the factor 2 that their own preceding display carries on the cross terms. Strengthening Assumption 3.4 to $s_0 \geq (4\nu^{-2}\omega(X) + 1)s^*$, and Assumption 3.5 so that the noise budget is $A/8$ rather than $A/4$, repairs this with room to spare; see (116) below. Nothing else in their argument, or in ours, is affected.

LEMMA A.5 (Proxy bound). *For every $\gamma \in \mathcal{M}(s)$ and every $j \in \gamma \setminus \gamma^*$,*

$$(97) \quad \|(\Phi_\gamma - \Phi_{\gamma \setminus \{j\}})X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 \leq n\omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2.$$

PROOF. This is the chain of displays in the proof of Lemma A.4b, run with $\tilde{\gamma}$ replaced by γ and k_γ by j . Every step used only $j \notin \gamma^*$ and $\gamma \in \mathcal{M}(s)$: first $\Phi_\gamma X_{\gamma^* \cap \gamma} = \Phi_{\gamma \setminus \{j\}} X_{\gamma^* \cap \gamma}$, so the left side equals $\|(\Phi_\gamma - \Phi_{\gamma \setminus \{j\}}) X_{\gamma^* \setminus \gamma} \beta_{\gamma^* \setminus \gamma}^*\|_2^2$; then the identification with a difference of least-squares residuals; then feasibility of zeroing the j th coordinate, which gives the bound $n|\hat{\beta}_j(\gamma)|^2$; then $\|\hat{\beta}(\gamma)\|_2^2 \leq \omega(X)\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2$. Taking $\gamma \supseteq \gamma^*$ makes the right side vanish and recovers the identity used in Section A.4. $\square$

A.3. Counting and paths.

LEMMA A.6 (Counting precedents). *Let $\bar{\eta} \in \mathcal{M}(s_0)$.*

- (a) $\mathcal{G}$ maps overfitted states to states containing γ^* . Consequently, if γ is underfitted then every $\bar{\gamma} \in \Lambda(\gamma)$ is underfitted.
- (b) $\bar{\eta}$ has at most $(s^* + 1)p$ immediate precedents that are underfitted — that is, states $\bar{\gamma} \neq \bar{\eta}$ with $\mathcal{G}(\bar{\gamma}) = \bar{\eta}$ and $\bar{\gamma}$ underfitted — and at most p that are overfitted.
- (c) Hence, for every $k \geq 0$, the number of $\bar{\gamma}$ with $\mathcal{G}^k(\bar{\gamma}) = \bar{\eta}$ and $\bar{\gamma}$ underfitted is at most $\{(s^* + 1)p\}^k$; and if $\bar{\eta} \supseteq \gamma^*$, the number of overfitted $\bar{\gamma} \in \Lambda_k(\bar{\eta})$ is at most p^k .

PROOF. **a** An overfitted $\bar{\gamma}$ is treated by rule (G1), which deletes $\ell_{\bar{\gamma}} \in \bar{\gamma} \setminus \gamma^*$ and therefore leaves $\gamma^* \subseteq \mathcal{G}(\bar{\gamma})$; and γ^* is a fixed point. So $\mathcal{G}^k(\bar{\gamma}) \supseteq \gamma^*$ for every $k \geq 0$, and no such state is underfitted.

b Let $\bar{\gamma} \neq \bar{\eta}$ be underfitted with $\mathcal{G}(\bar{\gamma}) = \bar{\eta}$. If rule (G2) applies then $\bar{\eta} = \bar{\gamma} \cup \{j_{\bar{\gamma}}\}$ with $j_{\bar{\gamma}} \in \gamma^*$, so $\bar{\gamma} = \bar{\eta} \setminus \{j\}$ for some $j \in \gamma^* \cap \bar{\eta}$: at most s^* states. If rule (G3) applies then $\bar{\eta} = \bar{\gamma} \cup \{j_{\bar{\gamma}}\} \setminus \{k_{\bar{\gamma}}\}$ with $j_{\bar{\gamma}} \in \gamma^*$ and $k_{\bar{\gamma}} \in S^c$, so $\bar{\gamma} = \bar{\eta} \cup \{k\} \setminus \{j\}$ with $j \in \gamma^* \cap \bar{\eta}$ and $k \in S^c$: at most $s^*(p - s^*)$ states. The total is at most $s^*(1 + p - s^*) \leq s^*(1 + p) \leq (s^* + 1)p$, the last step because $s^* \leq p$. If instead $\bar{\gamma}$ is overfitted then rule (G1) applies and $\bar{\gamma} = \bar{\eta} \cup \{\ell\}$ with $\ell \in S^c$: at most p states.

c Iterate **b**: a state counted at level k is an immediate precedent, of the stated type, of one counted at level $k - 1$. For the second claim, **a** shows that an overfitted $\bar{\gamma} \in \Lambda_k(\bar{\eta})$ reaches $\bar{\eta}$ through states containing γ^* only, so each of the k steps deletes an index of S^c ; hence $\bar{\gamma} = \bar{\eta} \cup u$ with $u \subseteq S^c$ and $|u| = k$, and there are at most $\binom{p}{k} \leq p^k$ such u . $\square$

LEMMA A.7 (Path properties; YWJ, Lemma 2).

- (a) $\mathcal{G}^k(\gamma) = \gamma^*$ for some $k \leq d_H(\gamma, \gamma^*)$, every $\gamma \in \mathcal{M}(s_0)$; in particular $\Lambda(\gamma^*) = \mathcal{M}(s_0)$.
- (b) $|T_{\gamma, \gamma^*}| \leq d_H(\gamma, \gamma^*) \leq s_0 + s^* \leq 2s_0$, and $\ell(\mathcal{T}) \leq 4s_0$.
- (c) For a directed edge $e = (\bar{\gamma}, \mathcal{G}(\bar{\gamma}))$,

$$\{(\gamma, \gamma') : T_{\gamma, \gamma'} \ni e\} \subseteq \Lambda(\bar{\gamma}) \times \mathcal{M}(s_0).$$

PROOF. **a** By the second remark after Definition 3.3, $d_H(\mathcal{G}(\gamma), \gamma^*) < d_H(\gamma, \gamma^*)$ whenever $\gamma \neq \gamma^*$, so the orbit reaches the unique fixed point γ^* in at most $d_H(\gamma, \gamma^*)$ steps. Hence every state lies in $\Lambda(\gamma^*)$.

b Each step of $\mathcal{G}$ decreases $d_H(\cdot, \gamma^*)$ by at least one, so the orbit has at most $d_H(\gamma, \gamma^*)$ edges; and $d_H(\gamma, \gamma^*) \leq |\gamma| + |\gamma^*| \leq s_0 + s^*$, which is at most $2s_0$ because $s^* < s_0$. Since $T_{\gamma, \gamma'}$ is contained in the concatenation of T_{γ, γ^*} with the reverse of T_{γ', γ^*} , its length is at most $4s_0$.

c By construction $T_{\gamma, \gamma'}$ traverses its first segment in the direction of $\mathcal{G}$ and its second segment against it. The edge e points in the direction of $\mathcal{G}$, so it can only occur in the first segment, which is part of T_{γ, γ^*} . Hence $\bar{\gamma} \in T_{\gamma, \gamma^*}$, that is $\gamma \in \Lambda(\bar{\gamma})$, while γ' is unconstrained. $\square$

A.4. Proof of Theorem 4.1: overfitted states. Throughout Sections A.4 to A.6 we write $\gamma' := \mathcal{G}(\gamma)$ and start from (8) with the size change read off from (25).

Here $\gamma \supseteq \gamma^*$, $\gamma \neq \gamma^*$, and rule (G1) applies: $\gamma' = \gamma \setminus \{\ell_\gamma\}$ with $\ell_\gamma \in \gamma \setminus \gamma^*$, so $|\gamma'| - |\gamma| = -1$ and (8) gives

$$(98) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = -\kappa \log p - \lambda \alpha \log p + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2}.$$

Since $\gamma' \subset \gamma$ and $|\gamma| - |\gamma'| = 1$, the matrix $\Phi_\gamma - \Phi_{\gamma'}$ is an orthogonal projection of rank one, and

$$(99) \quad \|Y\|_2^2 (R_\gamma^2 - R_{\gamma'}^2) = Y^\top (\Phi_\gamma - \Phi_{\gamma'}) Y = \|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2 \geq 0,$$

so that, using $\log(1+x) \leq x$ and then Lemma A.1,

$$(100) \quad \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} = \frac{n}{2} \log \left(1 + \frac{\|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2}{\|Y\|_2^2 (g^{-1} + 1 - R_\gamma^2)} \right) \leq \frac{n}{2} \cdot \frac{8 \|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2}{n \sigma_0^2}.$$

It remains to bound $\|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2$. Since $\ell_\gamma \notin \gamma^*$ and $\gamma^* \subseteq \gamma$, we have $\gamma^* \subseteq \gamma'$, so $X_{\gamma^*} \beta_{\gamma^*}$ lies in the range of $\Phi_{\gamma'}$; and $\Phi_\gamma - \Phi_{\gamma'}$ annihilates that range, because for v in it $\Phi_\gamma v = v$, the range of $\Phi_{\gamma'}$ being contained in that of Φ_γ , so that $(\Phi_\gamma - \Phi_{\gamma'}) v = v - v = 0$. Hence, by (81),

$$(101) \quad (\Phi_\gamma - \Phi_{\gamma'}) Y = (\Phi_\gamma - \Phi_{\gamma'}) \tilde{w}, \quad \text{so} \quad \|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2 \leq 2(L + \tilde{L}) \sigma_0^2 \log p$$

by Lemma A.2b, applied to the nested pair $\gamma' \subset \gamma$ of $\mathcal{M}(s_0) \subseteq \mathcal{M}(s)$ with $|\gamma| - |\gamma'| = 1$. Substituting (101) into (100),

$$(102) \quad \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} \leq 8(L + \tilde{L}) \log p,$$

and substituting this into (98), using $\lambda \geq 0$ so that multiplying (102) by λ preserves the inequality,

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} \leq -\kappa \log p - \lambda \{ \alpha - 8(L + \tilde{L}) \} \log p,$$

which is (27). $\square$

A.5. Proof of Theorem 4.1: underfitted, unsaturated states. Here $\gamma^* \setminus \gamma \neq \emptyset$ and $|\gamma| < s_0$, so rule (G2) applies: $\gamma' = \gamma \cup \{j_\gamma\}$, $|\gamma'| - |\gamma| = +1$, and (8) gives

$$(103) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = \kappa \log p + \lambda \alpha \log p + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2}.$$

Note $s^* \geq 1$, since $\gamma^* \setminus \gamma \neq \emptyset$. The matrix $\Phi_{\gamma'} - \Phi_\gamma$ is an orthogonal projection of rank one, and as in (99),

$$(104) \quad \|Y\|_2^2 (R_{\gamma'}^2 - R_\gamma^2) = \|(\Phi_{\gamma'} - \Phi_\gamma) Y\|_2^2,$$

so that, by $\log(1+x) \leq x$ applied with $x = -\|(\Phi_{\gamma'} - \Phi_\gamma) Y\|_2^2 / \{\|Y\|_2^2 (g^{-1} + 1 - R_\gamma^2)\} \in (-1, 0]$,

$$(105) \quad \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} \leq -\frac{n}{2} \cdot \frac{\|(\Phi_{\gamma'} - \Phi_\gamma) Y\|_2^2}{\|Y\|_2^2 (g^{-1} + 1 - R_\gamma^2)}.$$

Step 1: the numerator. By Lemma A.4a, $\|(\Phi_{\gamma'} - \Phi_{\gamma})X_{\gamma^*}\beta_{\gamma^*}^*\|_2 \geq A$, and by Lemma A.2b applied to the nested pair $\gamma \subset \gamma'$, together with (87),

$$(106) \quad \|(\Phi_{\gamma'} - \Phi_{\gamma})\tilde{w}\|_2^2 \leq \frac{n\nu^2 C_{\beta}^2}{64} \leq \frac{A^2}{64}, \quad \text{that is} \quad \|(\Phi_{\gamma'} - \Phi_{\gamma})\tilde{w}\|_2 \leq \frac{A}{8}.$$

Hence, by the triangle inequality and (81),

$$(107) \quad \|(\Phi_{\gamma'} - \Phi_{\gamma})Y\|_2 \geq A - \frac{A}{8} = \frac{7A}{8} > 0, \quad \text{so} \quad \|(\Phi_{\gamma'} - \Phi_{\gamma})Y\|_2^2 \geq \frac{49}{64}A^2 \geq \frac{1}{2}A^2.$$

Step 2: the denominator. By Lemma A.3 and the definition (83), which gives $\|(I - \Phi_{\gamma})X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 = |\gamma^* \setminus \gamma|A^2/\nu \leq s^*A^2/\nu$,

$$(108) \quad \|Y\|_2^2(g^{-1} + 1 - R_{\gamma}^2) \leq \frac{2s^*A^2}{\nu} + 8n\sigma_0^2.$$

Step 3: combining. By (107) and (108), then by the third inequality of (82) with $a = 2s^*A^2/\nu > 0$ and $b = 8n\sigma_0^2 > 0$,

$$(109) \quad \frac{n}{2} \cdot \frac{\|(\Phi_{\gamma'} - \Phi_{\gamma})Y\|_2^2}{\|Y\|_2^2(g^{-1} + 1 - R_{\gamma}^2)} \geq \frac{n}{4} \cdot \frac{A^2}{2s^*A^2/\nu + 8n\sigma_0^2} = \frac{n\nu}{8s^*} \cdot \frac{a}{a+b} \geq \frac{n\nu}{8s^*} \min\left\{\frac{1}{2}, \frac{s^*A^2}{8\nu n\sigma_0^2}\right\},$$

that is, distributing $n\nu/(8s^*)$ over the minimum and then using $A^2 \geq n\nu^2 C_{\beta}^2$ from (87),

$$(110) \quad \frac{n}{2} \cdot \frac{\|(\Phi_{\gamma'} - \Phi_{\gamma})Y\|_2^2}{\|Y\|_2^2(g^{-1} + 1 - R_{\gamma}^2)} \geq \min\left\{\frac{n\nu}{16s^*}, \frac{A^2}{64\sigma_0^2}\right\} \geq \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_{\beta}^2}{64\sigma_0^2}\right\}.$$

Combining (103), (105) and (110), and using $\lambda \geq 0$,

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} \leq \kappa \log p + \lambda \left\{ \alpha \log p - \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_{\beta}^2}{64\sigma_0^2}\right\} \right\},$$

which is (28). $\square$

A.6. Proof of Theorem 4.1: underfitted, saturated states. Here $\gamma^* \setminus \gamma \neq \emptyset$ and $|\gamma| = s_0$, so rule (G3) applies: with $\tilde{\gamma} = \gamma \cup \{j_{\gamma}\}$ and $\gamma' = \tilde{\gamma} \setminus \{k_{\gamma}\}$ we have $|\gamma'| = |\gamma|$, and (8) gives

$$(111) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_{\gamma}^2},$$

the inclusion penalty having cancelled identically. Note that $\tilde{\gamma} \in \mathcal{M}(s)$, since $|\tilde{\gamma}| = s_0 + 1 \leq s$, and that both γ and γ' are subsets of $\tilde{\gamma}$ of cardinality $|\tilde{\gamma}| - 1$, so that $\Phi_{\tilde{\gamma}} - \Phi_{\gamma}$ and $\Phi_{\tilde{\gamma}} - \Phi_{\gamma'}$ are orthogonal projections of rank one. Following YWJ, set

$$(112) \quad v_1 := (\Phi_{\tilde{\gamma}} - \Phi_{\gamma})X_{\gamma^*}\beta_{\gamma^*}^*, \quad v_2 := (\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^*}\beta_{\gamma^*}^*.$$

Adding and subtracting $Y^{\top} \Phi_{\tilde{\gamma}} Y$,

$$(113) \quad \|Y\|_2^2(R_{\gamma'}^2 - R_{\gamma}^2) = Y^{\top} \Phi_{\gamma'} Y - Y^{\top} \Phi_{\gamma} Y = \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})Y\|_2^2 - \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})Y\|_2^2.$$

By Lemma A.4a and b, $\|v_1\|_2 \geq A$ and $\|v_2\|_2 \leq A/2$; and by Lemma A.2b applied to each of the two nested pairs $\gamma \subset \tilde{\gamma}$ and $\gamma' \subset \tilde{\gamma}$ of $\mathcal{M}(s)$, exactly as in (106),

$$(114) \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})\tilde{w}\|_2 \leq \frac{A}{8}, \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})\tilde{w}\|_2 \leq \frac{A}{8}.$$

Therefore, by the triangle inequality applied to $(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})Y = v_1 + (\Phi_{\tilde{\gamma}} - \Phi_{\gamma})\tilde{w}$ and to $(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})Y = v_2 + (\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})\tilde{w}$,

$$(115) \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})Y\|_2 \geq A - \frac{A}{8} = \frac{7A}{8} > 0, \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})Y\|_2 \leq \frac{A}{2} + \frac{A}{8} = \frac{5A}{8},$$

and squaring, which is legitimate because both bounds are non-negative,

$$(116) \quad \|Y\|_2^2(R_{\gamma'}^2 - R_{\gamma}^2) \geq \frac{49}{64}A^2 - \frac{25}{64}A^2 = \frac{24}{64}A^2 \geq \frac{A^2}{4} > 0.$$

Inequality (105) holds verbatim, with $\|(\Phi_{\gamma'} - \Phi_{\gamma})Y\|_2^2$ replaced by the left-hand side of (116), and repeating Steps 2 and 3 of Section A.5 with $A^2/2$ replaced by $A^2/4$ in the numerator of (109) halves the right-hand side of (110):

$$\frac{n}{2} \cdot \frac{\|Y\|_2^2(R_{\gamma'}^2 - R_{\gamma}^2)}{\|Y\|_2^2(g^{-1} + 1 - R_{\gamma}^2)} \geq \frac{n}{8} \cdot \frac{A^2}{2s^*A^2/\nu + 8n\sigma_0^2} \geq \frac{1}{2} \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2C_{\beta}^2}{64\sigma_0^2}\right\}.$$

Substituting into (111) and using $\lambda \geq 0$ gives the third bound (29). $\square$

A.7. Proofs for Section 3.

PROOF OF LEMMA 3.2. Write $q(\gamma, \gamma')$ for the probability that $\tilde{\mathbf{P}}$ proposes γ' from γ . On $\mathcal{N}_1$ it equals $1/(2p)$, symmetric in (γ, γ') . On $\mathcal{N}_2$ the move preserves the size, so $|\gamma'| = |\gamma|$ and $q(\gamma, \gamma') = \{2|\gamma|(p - |\gamma|)\}^{-1} = q(\gamma', \gamma)$; in particular an exchange never leaves $\mathcal{M}(s_0)$. Hence q is symmetric, and for $\gamma \neq \gamma'$

$$(117) \quad \mu(\gamma) \tilde{\mathbf{P}}(\gamma, \gamma') = q(\gamma, \gamma') \min\{\mu(\gamma), \mu(\gamma')\},$$

which is symmetric; this is detailed balance for $\tilde{\mathbf{P}}$, hence for $\mathbf{P}$, and it shows $\mathbf{Q}$ is symmetric.

For irreducibility, recall that μ has full support on $\mathcal{M}(s_0)$. Given $\gamma \neq \gamma'$, delete the elements of γ one at a time and then add those of γ' one at a time; every intermediate state is a subset of γ or of γ' , hence lies in $\mathcal{M}(s_0)$, and consecutive states differ by one flip, so each transition has positive probability by (15). Aperiodicity is $\mathbf{P}(\gamma, \gamma) \geq 1/2 > 0$.

For non-negative definiteness, $\tilde{\mathbf{P}}$ is a reversible Markov kernel, hence self-adjoint on $\mathcal{L}^2(\mu)$ with spectrum in $[-1, 1]$, so $\mathbf{P} = \frac{1}{2}(\mathbf{I} + \tilde{\mathbf{P}})$ has spectrum in $[0, 1]$ and its absolute spectral gap coincides with $1 - \lambda_2$.

For (17), combine (16) and (117): $\mathbf{Q}(\gamma, \gamma') = \frac{1}{2}q(\gamma, \gamma') \min\{\mu(\gamma), \mu(\gamma')\}$, and $q(\gamma, \gamma') \geq 1/(2ps_0)$ in both cases: on $\mathcal{N}_1$ because $1/(2p) \geq 1/(2ps_0)$, and on $\mathcal{N}_2$ because $|\gamma| \leq s_0$ and $p - |\gamma| \leq p$. $\square$

A.8. Proofs for Section 4.

PROOF OF COROLLARY 4.1. **a** Collecting the two multiples of $\log p$ in (27) gives

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq -[\kappa + \lambda\{\alpha - 8(L + \tilde{L})\}] \log p = -[(1 - \lambda)\kappa + \lambda\{\kappa + \alpha - 8(L + \tilde{L})\}] \log p,$$

which is the first inequality of (30). The bracket is a convex combination of κ and $\kappa + \alpha - 8(L + \tilde{L})$, hence at least their minimum for every $\lambda \in [0, 1]$, and Assumption 3.3 bounds both below by 2: the first because $\kappa \geq 2$, the second because $\kappa + \alpha \geq 8(L + \tilde{L}) + 2$. This is the second inequality.

b For an underfitted unsaturated γ , the bound (28) is at most $-3 \log p$ precisely when $\lambda\{\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2C_{\beta}^2}{64\sigma_0^2}\} - \alpha \log p\} \geq (\kappa + 3) \log p$, that is when λ is at least the first term of (31),

the coefficient $\min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} - \alpha \log p$ being positive by hypothesis. For an underfitted saturated γ , the bound (29) is at most $-3 \log p$ precisely when $\lambda \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} \geq 6 \log p$, that is when λ is at least the second term of (31). Taking the maximum gives (32).

Identity (33) is (8) rearranged: its right-hand side is affine in λ and agrees with the left-hand side at $\lambda = 0$, where the fit term vanishes and the ratio is the prior ratio, and at $\lambda = 1$. $\square$

PROOF OF COROLLARY 4.2. Assumption 3.5 says exactly that $n\nu^2 C_\beta^2 / (64\sigma_0^2) \geq 2(L + \tilde{L} + \alpha + \kappa) \log p$, and Assumption 3.6 exactly that $n\nu / (16s^*) \geq 2(L + \tilde{L} + \alpha + \kappa) \log p$. The floor $\min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\}$ is the smaller of those two left-hand sides, which gives the first inequality of (34); the second holds because $2(L + \tilde{L} + \alpha + \kappa) > \alpha$, the quantities $L, \tilde{L}, \kappa$ being non-negative and $\alpha > 0$. In particular the standing hypothesis $\min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} > \alpha \log p$ of Corollary 4.1 holds and λ_* is finite.

Write $\Psi := \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\}$ and $Q := L + \tilde{L} + \alpha + \kappa$. By Assumptions 3.2 and 3.3, $L\nu \geq 4$ and $\nu \leq 1$ give $L \geq 4$, so $Q \geq 4 + \frac{1}{2} + 2 > 6$. By (31) and (11), the first term of the maximum is at most 1 if and only if $\Psi \geq (\kappa + \alpha + 3) \log p$, for which (34) asks $2Q \geq \kappa + \alpha + 3$, that is $2(L + \tilde{L}) + \alpha + \kappa \geq 3$; and the second term is at most 1 if and only if $\Psi \geq 6 \log p$, for which it asks $Q \geq 3$. Both hold, with room to spare, since $Q > 6$ and $2(L + \tilde{L}) + \alpha + \kappa \geq Q > 6$. This is (35). $\square$

PROOF OF COROLLARY 4.3. Write $\Psi := \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\}$, $Q := L + \tilde{L} + \alpha + \kappa$ and $B := L + \tilde{L}$, so that $2Q - \alpha = 2B + \alpha + 2\kappa$. By (34), $\Psi \geq 2Q \log p$, and therefore

$$(118) \quad \lambda_*(q) \leq \max\left\{\frac{\kappa + q}{2B + \alpha + 2\kappa}, \frac{q}{Q}\right\}.$$

As in the proof of Corollary 4.2, Assumption 3.2 gives $L \geq 4$ and hence $B \geq 4$, while Assumption 3.3 gives $\alpha \geq \frac{1}{2}$ and $\kappa \geq 2$, so $Q \geq \frac{13}{2}$.

The first term. Writing it as

$$\frac{\kappa + q}{2B + \alpha + 2\kappa} = \frac{1}{2} - \frac{2B + \alpha - 2q}{2(2B + \alpha + 2\kappa)},$$

it is strictly below $\frac{1}{2}$ as soon as $2B + \alpha > 2q$, which holds for every $q \leq 4$ because $2B + \alpha \geq \frac{17}{2}$. The subtracted term is positive and decreases to 0 as $\kappa \rightarrow \infty$, so the first term increases in κ with supremum $\frac{1}{2}$, not attained; and it decreases in B and in α , so the supremum is approached at $B = 4, \alpha = \frac{1}{2}$.

The second term. For the exponent $q = 3$ of (32), $q/Q \leq 6/13 < \frac{1}{2}$. For a general $q \leq 4$, Assumption 3.3 gives $\kappa + \alpha \geq 8B + 2 \geq 34$, hence $Q = B + (\kappa + \alpha) \geq 38$ and $q/Q \leq \frac{2}{19}$.

Both terms of (118) are therefore strictly below $\frac{1}{2}$, which is (36), and the first approaches $\frac{1}{2}$ along an admissible sequence: $L = 4, \nu = 1, \tilde{L} = 0$ satisfy Assumption 3.2, $\alpha = \frac{1}{2}$ and $\kappa \rightarrow \infty$ satisfy Assumption 3.3, and Assumptions 3.5 and 3.6 constrain only (n, ν, C_β, s^*) and are met by taking n large. Hence $\frac{1}{2}$ is the supremum.

Finally, the claim made after Corollary 4.3 about lowering the exponent: with p^{-q} in place of p^{-3} in (32), the underfitted series of (120) has ratio $a = (s^* + 1)p^{1-q}$, so $q = 2$ requires $(s^* + 1)p^{-1} \leq \frac{1}{2}$; by Assumption 3.4, $s^* + 1 \leq s \leq n/(256 \log p) \leq p/(256 \log p)$, which is far smaller than $p/2$. The overfitted series is untouched, and (43) becomes $\{1 - (s^* + 1)p^{-1}\}(1 - p^{-1})$. $\square$

PROOF OF COROLLARY 4.4. Write $a := (s^* + 1)p^{-2}$ and $b := p^{-1}$. Since $s^* \leq p$ we have $a \leq (p + 1)p^{-2} \leq \frac{1}{2}$ and $b \leq \frac{1}{2}$ for every $p \geq 3$.

Step 0: the drop along a path. If $\mathcal{G}^k(\bar{\gamma}) = \gamma$ then, telescoping and applying Corollary 4.1 at each of the k source states,

$$(119) \quad \frac{\pi_n^{(\lambda)}(\bar{\gamma} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} = \prod_{i=0}^{k-1} \frac{\pi_n^{(\lambda)}(\mathcal{G}^i(\bar{\gamma}) | Y)}{\pi_n^{(\lambda)}(\mathcal{G}^{i+1}(\bar{\gamma}) | Y)} \leq p^{-3k_u - 2k_o},$$

where k_u and k_o count the underfitted and the overfitted source states among $\bar{\gamma}, \dots, \mathcal{G}^{k-1}(\bar{\gamma})$, so that $k_u + k_o = k$: part **b** of that corollary applies at the first and part **a** at the second, and every state other than γ^* is one or the other. In the two uses below the sources are of a single kind, so the exponent is $3k$ or $2k$ throughout.

Step 1: γ underfitted. By Lemma A.6a every $\bar{\gamma} \in \Lambda(\gamma)$ is underfitted, so Lemma A.6c gives $|\Lambda_k(\gamma)| \leq \{(s^* + 1)p\}^k$, and with (119)

$$(120) \quad \frac{\pi_n^{(\lambda)}[\Lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \sum_{k \geq 0} \{(s^* + 1)p\}^k p^{-3k} = \sum_{k \geq 0} a^k = \frac{1}{1 - a},$$

the series converging because $a \leq \frac{1}{2}$.

Step 2: $\gamma \supseteq \gamma^$.* Split $\Lambda(\gamma) = \mathbb{M}_1 \sqcup \mathbb{M}_2$ into the precedents that contain γ^* and those that do not, so that $\gamma \in \mathbb{M}_1$ and every member of $\mathbb{M}_2$ is underfitted. By Lemma A.6c and (119),

$$(121) \quad \frac{\pi_n^{(\lambda)}[\mathbb{M}_1 | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \sum_{k \geq 0} p^k p^{-2k} = \frac{1}{1 - b}.$$

For $\bar{\gamma} \in \mathbb{M}_2$ let $f(\bar{\gamma})$ be the first state on the path $\bar{\gamma} \rightarrow \mathcal{G}(\bar{\gamma}) \rightarrow \dots \rightarrow \gamma$ that contains γ^* ; it exists because γ itself does, and it lies in $\mathbb{M}_1$ because it is a precedent of γ . Every state strictly before $f(\bar{\gamma})$ on that path is underfitted, by the definition of f , so Lemma A.6c applies to the chain from $\bar{\gamma}$ to $f(\bar{\gamma})$, whose length is at least 1 because $\bar{\gamma} \notin \mathbb{M}_1$. Grouping the sum over $\mathbb{M}_2$ by the value of f , and using (119) for the inner ratios,

$$(122) \quad \begin{aligned} \frac{\pi_n^{(\lambda)}[\mathbb{M}_2 | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} &= \sum_{\bar{\gamma} \in \mathbb{M}_1} \frac{\pi_n^{(\lambda)}(\bar{\gamma} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} \sum_{\bar{\gamma}: f(\bar{\gamma}) = \bar{\gamma}} \frac{\pi_n^{(\lambda)}(\bar{\gamma} | Y)}{\pi_n^{(\lambda)}(\bar{\gamma} | Y)} \\ &\leq \frac{\pi_n^{(\lambda)}[\mathbb{M}_1 | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \cdot \sum_{k \geq 1} a^k \leq \frac{1}{1 - b} \cdot \frac{a}{1 - a}. \end{aligned}$$

Adding (121) and (122),

$$\frac{\pi_n^{(\lambda)}[\Lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \frac{1}{1 - b} \left(1 + \frac{a}{1 - a} \right) = \frac{1}{(1 - a)(1 - b)},$$

which is the middle expression in (38). It dominates the bound $1/(1 - a)$ of Step 1, so both cases are covered by it, and every $\gamma \in \mathcal{M}(s_0)$ falls into one of them: a state other than γ^* is either underfitted or overfitted, and $\gamma^* \supseteq \gamma^*$. Finally $a, b \leq \frac{1}{2}$ gives $\{(1 - a)(1 - b)\}^{-1} \leq 4$. $\square$

PROOF OF PROPOSITION 4.1. Identity (39) is (9) with $j = j_\gamma$, read in the direction $\log\{\pi_n^{(\lambda)}(\gamma | Y)/\pi_n^{(\lambda)}(\gamma' | Y)\} = -\log\{\pi_n^{(\lambda)}(\gamma' | Y)/\pi_n^{(\lambda)}(\gamma | Y)\}$. The right-hand side of (39) is non-positive if and only if $\lambda \log\{\mathcal{L}_n(Y | \mathcal{G}(\gamma))/\mathcal{L}_n(Y | \gamma)\} \geq \kappa \log p$, which gives the stated equivalence; at $\lambda = 0$ it equals $\kappa \log p > 0$; and if the likelihood ratio is at most

one then its logarithm is non-positive, so for $\lambda \geq 0$ the right-hand side is at least $\kappa \log p > 0$. The characterisation of λ_{crit} follows by taking the maximum over the finitely many underfitted unsaturated γ , with the convention that a state whose likelihood ratio is at most one contributes $+\infty$.

For (41), form the ratio of two copies of (6) with $\gamma' = \gamma \cup \{j_\gamma\}$:

$$\begin{aligned} \log \frac{\mathcal{L}_n(Y | \gamma')}{\mathcal{L}_n(Y | \gamma)} &= -\alpha \log p - \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} \\ &\geq -\alpha \log p + \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\}, \end{aligned}$$

the last step by (105) and (110). The right-hand side being positive by hypothesis, the quotient defining λ_{crit} is at most $\kappa \log p / \{\min \{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \} - \alpha \log p\}$ for every such γ , and taking the maximum gives the claim. $\square$

A.9. Proofs for Section 5.

PROOF OF LEMMA 5.1. By Corollary 4.1 the drop bound gives $\pi_n^{(\lambda)}(\bar{\gamma} | Y) \leq \pi_n^{(\lambda)}(\mathcal{G}(\bar{\gamma}) | Y)$ at every $\bar{\gamma} \neq \gamma^*$ — each such state being either overfitted or underfitted, and both cases being covered by its hypotheses — so the inner maximum in (21) equals one, and Corollary 4.4 bounds the remaining factor by 4. Hence $\rho(\mathcal{T}) \leq 4ps_0 \cdot 4 = 16ps_0$. With $\ell(\mathcal{T}) \leq 4s_0$ from Lemma A.7b, Sinclair's bound (19) gives $\text{Gap}(\mathbf{P}_\lambda) \geq (16ps_0 \cdot 4s_0)^{-1}$, which is (42); it is the absolute spectral gap by Lemma 3.2. $\square$

PROOF OF PROPOSITION 5.1. *Step 1: the precedents of γ^* are everything.* By Lemma A.7a, every $\bar{\gamma} \in \mathcal{M}(s_0)$ satisfies $\mathcal{G}^k(\bar{\gamma}) = \gamma^*$ for some $k \geq 0$, which is the defining condition (20) for membership of $\Lambda(\gamma^*)$. Hence $\Lambda(\gamma^*) \supseteq \mathcal{M}(s_0)$, and the reverse inclusion holds by (20), so $\Lambda(\gamma^*) = \mathcal{M}(s_0)$. Since $\pi_n^{(\lambda)}(\cdot | Y)$ is a probability measure on $\mathcal{M}(s_0)$ by (7),

$$(123) \quad \pi_n^{(\lambda)}[\Lambda(\gamma^*) | Y] = \pi_n^{(\lambda)}[\mathcal{M}(s_0) | Y] = 1.$$

Step 2: Corollary 4.4 applies at $\gamma = \gamma^$.* That corollary is stated for every $\gamma \in \mathcal{M}(s_0)$, and γ^* is covered by Step 2 of its proof, which treats every γ with $\gamma \supseteq \gamma^*$; the case $\gamma = \gamma^*$ is admissible there because $\gamma^* \supseteq \gamma^*$. Substituting (123) into (38) therefore gives

$$(124) \quad \frac{1}{\pi_n^{(\lambda)}(\gamma^* | Y)} = \frac{\pi_n^{(\lambda)}[\Lambda(\gamma^*) | Y]}{\pi_n^{(\lambda)}(\gamma^* | Y)} \leq \frac{1}{\{1 - (s^* + 1)p^{-2}\}\{1 - p^{-1}\}}.$$

Step 3: inverting. Put $u := \{1 - (s^* + 1)p^{-2}\}\{1 - p^{-1}\}$, the right-hand side of (43). As in the proof of Corollary 4.4, $p \geq 3$ gives $(s^* + 1)p^{-2} \leq \frac{1}{2}$ and $p^{-1} \leq \frac{1}{2}$, so both factors of u lie in $[\frac{1}{2}, 1]$ and hence

$$(125) \quad \frac{1}{4} \leq u \leq 1.$$

Also $\pi_n^{(\lambda)}(\gamma^* | Y) \in (0, 1]$ by (7). Inequality (124) therefore reads

$$\frac{1}{\pi_n^{(\lambda)}(\gamma^* | Y)} \leq \frac{1}{u},$$

an inequality between two elements of $(0, \infty)$. The map $x \mapsto 1/x$ is strictly decreasing on $(0, \infty)$, so applying it to both sides reverses the inequality and gives $\pi_n^{(\lambda)}(\gamma^* | Y) \geq u$, which is the first inequality of (43). The second is the left-hand inequality of (125). $\square$

PROOF OF LEMMA 5.2. Apply (8) with $\gamma' = \gamma^*$:

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma^* | Y)} = (|\gamma^*| - |\gamma|)(\kappa + \lambda\alpha) \log p + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma^*}^2}{g^{-1} + 1 - R_{\gamma}^2}.$$

For the first term, $|\gamma^*| - |\gamma| \geq -|\gamma| \geq -s_0$ and the brace is non-negative, so it is at least $-s_0\{\kappa \log p + \lambda\alpha \log p\}$. For the second, $R_{\gamma^*}^2 \leq 1$ and $R_{\gamma}^2 \geq 0$ give

$$\frac{g^{-1} + 1 - R_{\gamma^*}^2}{g^{-1} + 1 - R_{\gamma}^2} \geq \frac{g^{-1}}{g^{-1} + 1} = \frac{1}{1 + g},$$

so it is at least $-\lambda n \alpha \log p$. Adding the two and using $\lambda \geq 0$ gives (44). For (45), combine it with $\pi_n^{(\lambda)}(\gamma^* | Y) \geq \frac{1}{4}$ from Proposition 5.1. $\square$

PROOF OF THEOREM 5.1. Lemma 3.2 makes $\mathbf{P}_\lambda$ lazy and reversible, so (26) applies with $\mu = \pi_n^{(\lambda)}(\cdot | Y)$. Substitute the gap bound of Lemma 5.1 and the mass bound (45), whose logarithm is $\kappa s_0 \log p + \lambda(n + s_0)\alpha \log p + \log 4$. $\square$

A.10. Proofs for Section 6.

PROOF OF LEMMA 6.1. By (47), $1 + g(1 - R_{\gamma}^2) = g D_{\gamma} / \|Y\|_2^2$, so by (5) and (6) the unnormalised mass $\pi(\gamma) \mathcal{L}_n(Y | \gamma)^\lambda$ is $p^{-\kappa|\gamma|} (1 + g)^{-\lambda|\gamma|/2} D_{\gamma}^{-\lambda n/2}$ times factors free of γ , which cancel in the normalisation; (11) turns this into $w_\lambda(\gamma)$. The second equality of (49) is the same computation at $|\gamma'| - |\gamma| = 1$. $\square$

PROOF OF LEMMA 6.2. Let $\mathbf{T}$ be the block Gibbs sampler for the partition $\{S\} \cup \{\{j\}\}_{j \in S^c}$ of the coordinates: it picks one of the $p - s^* + 1$ blocks uniformly and resamples it from its conditional law. Its Dirichlet form is $\mathcal{E}_{\mathbf{T}}(f, f) = (p - s^* + 1)^{-1} [\sum_{j \in S^c} \mathbb{E} \text{Var}(f | \gamma_{-j}) + \mathbb{E} \text{Var}(f | \gamma_{S^c})]$, so (55) is exactly the statement $\text{Gap}(\mathbf{T}) \geq C_p^{-1} (p - s^* + 1)^{-1}$, and it suffices to bound the Dobrushin interdependence matrix R of the block dynamics, whose entry $R_{BB'}$ is the largest total-variation change in the conditional law of block B caused by changing block B' alone.

Two estimates. First, for $B = \{j\}$ with $j \in S^c$: by (52) the conditional law of γ_j is Bernoulli with parameter at most p^{-2} *whatever* the other coordinates, so any two such laws are within p^{-2} in total variation and $R_{\{j\}B'} \leq p^{-2}$ for every B' . Second, for $B = S$ we use only the trivial $R_{SB'} \leq 1$.

So R vanishes except on the row and column of S , where it is at most 1 and at most p^{-2} respectively. For such a matrix $\|R\|_{\text{op}}^2 \leq \|R^2\|_{\text{op}}$ and R^2 is supported on the two diagonal entries, with $(R^2)_{SS} \leq \sum_{j \in S^c} R_{S\{j\}} R_{\{j\}S} \leq (p - s^*) \cdot 1 \cdot p^{-2} \leq p^{-1}$ and $(R^2)_{\{j\}\{j\}} \leq R_{\{j\}S} R_{S\{j\}} \leq p^{-2}$. Hence $\|R\|_{\text{op}} \leq \sqrt{2/p} < 1$ for $p \geq 9$, and Dobrushin's condition for block dynamics gives $\text{Gap}(\mathbf{T}) \geq (1 - \|R\|_{\text{op}})/(p - s^* + 1)$, which is (55) with $C_p = (1 - \sqrt{2/p})^{-1}$. $\square$

PROOF OF THEOREM 6.1. Bound the two terms of (55) by the chain's own Dirichlet form. For $j \in S^c$, the flip $\gamma \mapsto \gamma^j$ is proposed with probability $1/(2p)$ and accepted by the Metropolis rule, and the chain is lazy, so $\mathbf{Q}_\lambda(\gamma, \gamma^j) = \min\{\pi_n^{(\lambda)}(\gamma), \pi_n^{(\lambda)}(\gamma^j)\}/(4p)$ exactly. Since $\mathbb{E} \text{Var}(f | \gamma_{-j}) = \sum_{\gamma_{-j}} \pi_n^{(\lambda)}(\gamma_{-j}) \text{Var}(f | \gamma_{-j})$ and a two-point variance is $\pi(0 | \cdot) \pi(1 | \cdot) \{\nabla_j f\}^2 \leq \min\{\pi(0 | \cdot), \pi(1 | \cdot)\} \{\nabla_j f\}^2$, summing over j gives $\sum_{j \in S^c} \mathbb{E} \text{Var}(f | \gamma_{-j}) \leq 4p \mathcal{E}_{\mathbf{P}_\lambda}(f, f)$.

For the second term, $\mathbb{E}\text{Var}(f \mid \gamma_{S^c}) = \sum_{\sigma} \pi_n^{(\lambda)}(F_{\sigma}) \text{Var}_{\mu_{\sigma}}(f)$, and by the definition (54) of $\kappa_S(\lambda)$ each $\text{Var}_{\mu_{\sigma}}(f) \leq \kappa_S(\lambda)^{-1} \mathcal{E}_{\mathbf{P}_{\lambda}|_{F_{\sigma}}}(f, f)$; the restricted forms sum to at most $\mathcal{E}_{\mathbf{P}_{\lambda}}(f, f)$, the edges within the fibres being disjoint and carrying the same weights as in the full chain. So the second term is at most $\kappa_S(\lambda)^{-1} \mathcal{E}_{\mathbf{P}_{\lambda}}(f, f)$, and (55) gives $\text{Var}(f) \leq C_p \{4p + 1/\kappa_S(\lambda)\} \mathcal{E}_{\mathbf{P}_{\lambda}}(f, f)$, which is (56); the second form follows from $a + b \leq 2 \max\{a, b\}$. $\square$

PROOF OF PROPOSITION 6.1. Fix σ and work inside F_{σ} , a copy of $\{A \subseteq S : |A| \leq s_0 - |\sigma|\}$ carrying μ_{σ} . The available moves are flips of influential coordinates, proposed with probability $1/(2p)$, and exchanges between two of them, proposed with probability $\{2|\gamma|(p - |\gamma|)\}^{-1} \geq \{2s^*p\}^{-1}$ since the moving part has size at most s^* ; with the laziness, $\mathbf{Q}_{\lambda}(e) \geq \min\{\mu_{\sigma}(z), \mu_{\sigma}(z')\} \pi_n^{(\lambda)}(F_{\sigma}) / (4ps^*)$ for every edge of the fibre. Sinclair's bound (19) applied to $\mathbf{P}_{\lambda}|_{F_{\sigma}}$, whose stationary law is μ_{σ} , therefore gives (69).

For the second display, take the ascent of Definition 6.3 or the crossover of Definition 6.4 on 2^S , whichever has the smaller congestion. Under Assumption 6.2 the first has $\varrho_{\lambda} \leq C_{\lambda}$ by the argument below and $L_{\lambda} \leq 4s^*$; the second has $\varrho_{\lambda}^{\times} \leq M_{\lambda}$ by Proposition 6.2 and $L_{\lambda}^{\times} \leq s^*$. Taking the better and bounding $L_{\lambda} \leq 4s^*$ in either case gives $\text{Gap}(\mathbf{P}_{\lambda}|_{F_{\sigma}}) \geq 1/\{16p(s^*)^2 \max(1, \min\{C_{\lambda}, M_{\lambda}\})\}$; the constant 8 of (70) is this with the crossover's $L_{\lambda}^{\times} \leq s^*$ and $\varrho \leq M_{\lambda}$, and the minimum over σ is $\kappa_S(\lambda)$.

Finally the congestion identity used above. For a memoryless uphill ensemble with a unique fixed point, the ordered pairs whose route traverses $(z, F_{\lambda}(z))$ in that direction are exactly those with

$$(126) \quad A \in \Lambda^{F_{\lambda}}(z) \quad \text{and} \quad A' \notin \Lambda^{F_{\lambda}}(z),$$

since the route follows the orbit of A until its first state common with that of A' . The demand is therefore $x_{\lambda}(z)\{1 - x_{\lambda}(z)\}$ exactly, and dividing by $\min\{\mu_{\sigma}(z), \mu_{\sigma}(F_{\lambda}(z))\} = \mu_{\sigma}(z)$, the step being uphill, gives $\varrho_{\lambda} \leq C_{\lambda}$ by (66). $\square$

PROOF OF PROPOSITION 6.2. Write $\gamma^* = \{j_1, \dots, j_{s^*}\}$ in the greedy order and $I_{<t} := \{j_1, \dots, j_{t-1}\}$, $I_{\geq t}$ for its complement in γ^* . Along the route (64) each coordinate is flipped at most once, so $L_{\lambda}^{\times} \leq s^*$ and the route is simple.

The encoding. Fix a directed edge $e = (z, z')$ of the route, flipping the coordinate j_t , and let (A, A') be a pair whose route traverses it, so that $z = z_{t-1}$ and $z' = z_t$. Put

$$(127) \quad \eta := (A \cap I_{<t}) \cup (A' \cap I_{\geq t}).$$

By (64), z agrees with A' on $I_{<t}$ and with A on $I_{\geq t}$, while η agrees with A on $I_{<t}$ and with A' on $I_{\geq t}$: the pair (z, η) is the pair (A, A') with the two halves exchanged across the cut between j_{t-1} and j_t . Hence

$$(128) \quad A = (\eta \cap I_{<t}) \cup (z \cap I_{\geq t}), \quad A' = (z \cap I_{<t}) \cup (\eta \cap I_{\geq t}),$$

so (A, A') is recovered from (e, η) and $(A, A') \mapsto \eta$ is injective on the pairs using e . Since $j_t \in I_{\geq t}$, η carries the A' -value at j_t , which is the value z' carries there; so the image of the encoding lies in the cylinder $H := \{\eta : \eta \cap \{j_t\} = z' \cap \{j_t\}\}$.

The congestion. The pairs using e can be read off directly. By (64), (A, A') traverses $e = (z_{t-1}, z_t)$ exactly when A agrees with z on $I_{\geq t}$ and A' agrees with z' on $I_{<t} \cup \{j_t\}$, the remaining coordinates of each being free. Writing $a := z \cap I_{<t}$, $b := z \cap I_{\geq t}$ and $b' := z' \cap I_{\geq t}$, the demand is therefore a product of two marginals,

$$(129) \quad \sum_{(A, A') \ni e} \bar{\pi}_{\lambda}(A) \bar{\pi}_{\lambda}(A') = \bar{\pi}_{\lambda}(b) \bar{\pi}_{\lambda}(a \cup \{j_t\} \cap z'),$$

the first factor being the mass of the states meeting $I_{\geq t}$ in b and the second that of the states meeting $I_{< t} \cup \{j_t\}$ as z' does. Bounding the second factor by the marginal of its prefix alone, it is at most $\bar{\pi}_\lambda(a)$, so

$$\frac{1}{\bar{\pi}_\lambda(z)} \sum_{(A, A') \ni e} \bar{\pi}_\lambda(A) \bar{\pi}_\lambda(A') \leq \frac{\bar{\pi}_\lambda(a) \bar{\pi}_\lambda(b)}{\bar{\pi}_\lambda(a \cup b)} \leq M_\lambda$$

by (67), the cut being at t . If instead $\bar{\pi}_\lambda(z') < \bar{\pi}_\lambda(z)$, bound the *first* factor of (129) by the marginal of $I_{> t}$ and divide by $\bar{\pi}_\lambda(z')$: the same computation with the cut at $t + 1$ gives M_λ again. Hence $\varrho_\lambda^\times \leq M_\lambda$.

The product defect. Let ν attain the minimum in (68) and write $\bar{\pi}_\lambda = \nu e^h$; replacing ν by the product measure proportional to $\nu e^{(\max h + \min h)/2}$ and renormalising changes h by a constant and leaves (68) unchanged, so we may assume $\max h = -\min h = r_\lambda/2$. Marginals inherit the bounds, $\bar{\pi}_\lambda(a) \leq e^{r_\lambda/2} \nu(a)$ and likewise for b , while $\bar{\pi}_\lambda(a \cup b) \geq e^{-r_\lambda/2} \nu(a) \nu(b)$ since ν is a product. Hence every ratio in (67) is at most $e^{3r_\lambda/2} \leq e^{2r_\lambda}$, which is the second inequality of (74); and $M_\lambda = 1$ when $r_\lambda = 0$, the marginals of a product measure factorising exactly. $\square$

PROOF OF PROPOSITION 6.3. Under Assumption 6.2 every step of F_λ is w_λ -increasing by Proposition 6.5, so every $A \in \Lambda^{F_\lambda}(z)$ has $w_\lambda(A) \leq w_\lambda(z)$ and hence $\bar{\pi}_\lambda(A) \leq \bar{\pi}_\lambda(z)$, the fibre measure being proportional to w_λ on its states; that is, $\Lambda^{F_\lambda}(z)$ is contained in the set defining $\sigma_\lambda(z)$ and $x_\lambda(z) \leq \sigma_\lambda(z)$. The map $u \mapsto u(1-u)$ increases on $[0, \frac{1}{2}]$ and is at most $\frac{1}{4}$ throughout, so $x_\lambda(z)\{1-x_\lambda(z)\} \leq \min\{\sigma_\lambda(z)(1-\sigma_\lambda(z)), \frac{1}{4}\}$ in either case, which is (75) after dividing by $\bar{\pi}_\lambda(z)$ and maximising.

For the two consequences: $\sigma_\lambda(z) \leq 1$ and $\bar{\pi}_\lambda(z) \geq e^{-1/p} \min_A \bar{\pi}_\lambda(A)$ give $K_\lambda \leq e^{1/p} 2^{s^*}$, the 2^{s^*} masses summing to one so the smallest is at most 2^{-s^*} . Under the separated spectrum, if $\sigma_\lambda(z) \leq \frac{1}{2}$ the ratio in (75) is at most $\sigma_\lambda(z)/\bar{\pi}_\lambda(z) \leq \theta$; and if $\sigma_\lambda(z) > \frac{1}{2}$ then $\theta \bar{\pi}_\lambda(z) \geq \sigma_\lambda(z) > \frac{1}{2}$, so $\bar{\pi}_\lambda(z) > 1/(2\theta)$ and the ratio is at most $\frac{1}{4} \cdot 2\theta = \theta/2$. $\square$

PROOF OF PROPOSITION 6.4. Linearity. By (48), $\log \pi_n^{(\lambda)}(A | Y) = -\frac{\lambda n}{2} \log D_A - (\kappa + \lambda\alpha)|A| \log p - \log Z$, and on a fibre $\log \bar{\pi}_\lambda(A)$ is this up to an additive constant. Given (a_0, a) , the measure ν with $\log \nu(A) = \text{const} - \frac{\lambda n}{2} \sum_{j \in A} a_j - (\kappa + \lambda\alpha)|A| \log p$ is a product measure, since its logarithm is affine in the indicators of A ; and

$$\log \frac{\bar{\pi}_\lambda(A)}{\nu(A)} = -\frac{\lambda n}{2} \left\{ \log D_A - a_0 - \sum_{j \in A} a_j \right\} + h_A + \text{const.}$$

Taking oscillations, $r_\lambda \leq \frac{\lambda n}{2} \delta + 1/p$ on choosing (a_0, a) optimal in (76). Conversely every product ν has $\log \nu$ affine in the indicators, so it is of the above form for some (a_0, a) once the size-linear part is absorbed, and the same identity gives $r_\lambda \geq \frac{\lambda n}{2} \delta - 1/p$. This is the first claim of Proposition 6.4.

The bound on δ . Write $D_\emptyset = (1 + g^{-1}) \|Y\|_2^2$, $Q_A := \|\Phi_A Y\|_2^2$ and $u_A := Q_A/D_\emptyset$, so that $D_A = D_\emptyset(1 - u_A)$ by (47) and $\log D_A = \log D_\emptyset + \log(1 - u_A)$. Take the trial fit $a_0 = \log D_\emptyset$ and $a_j = -q_j/D_\emptyset$, whose residual is

$$(130) \quad \log(1 - u_A) + \sum_{j \in A} \frac{q_j}{D_\emptyset} = \{\log(1 - u_A) + u_A\} + \frac{1}{D_\emptyset} \left\{ \sum_{j \in A} q_j - Q_A \right\}.$$

For the second bracket, let G_A be the principal submatrix of G indexed by A and $b_A := X_A^\top Y/n$, so that $Q_A = n b_A^\top G_A^{-1} b_A$ while $\sum_{j \in A} q_j = n \|b_A\|_2^2$, the columns being normalised.

Since $\|G_A - I\|_{\text{op}} \leq \|G - I\|_{\text{op}} = \varepsilon \leq \frac{1}{2}$, we have $\|G_A^{-1} - I\|_{\text{op}} = \|G_A^{-1}(I - G_A)\|_{\text{op}} \leq \varepsilon/(1 - \varepsilon) \leq 2\varepsilon$, whence

$$\left| \sum_{j \in A} q_j - Q_A \right| = n |b_A^\top (G_A^{-1} - I) b_A| \leq 2\varepsilon n \|b_A\|_2^2 = 2\varepsilon \sum_{j \in A} q_j \leq 2\varepsilon U D_\emptyset.$$

In particular $u_A \leq (1 + 2\varepsilon)U \leq 2U \leq \frac{1}{2}$. For the first bracket, $|\log(1 - u) + u| \leq u^2/\{2(1 - u)\} \leq u^2$ for $u \leq \frac{1}{2}$, so it is at most $4U^2$. Adding, the residual of (130) is at most $4U^2 + 2\varepsilon U$ in absolute value, and an oscillation is at most twice a supremum of absolute values, so $\delta \leq 8U^2 + 4\varepsilon U = 4U(2U + \varepsilon)$. The second half of (77) follows from (74) and the first claim. $\square$

PROOF OF PROPOSITION 6.5. By (48), $w_\lambda(\gamma)$ depends on γ only through D_γ and $|\gamma|$, and is decreasing in each.

Let $A \subseteq \gamma^*$ with $A \neq P_{m_\lambda}$, so that $V(A) > 0$ by the paragraph following (E2). We exhibit an admissible neighbour of A at which w_λ is not smaller; since $F_\lambda(A)$ maximises w_λ over the admissible neighbours, that proves $w_\lambda(F_\lambda(A)) \geq w_\lambda(A)$.

If $A \neq P_{|A|}$, take the exchange of (E1), which replaces a_1 by b_1 . It is admissible, and it leaves $|A|$ unchanged, so the tilt $-(\kappa + \lambda\alpha)|A| \log p$ of (48) does not change; by (B1) it does not increase D , so it does not decrease w_λ , and this for every $\lambda \geq 0$.

If $A = P_l$ with $l \neq m_\lambda$, take the flip of (E2), which is admissible and moves l one step towards m_λ . The increments $\varphi_\lambda(m + 1) - \varphi_\lambda(m) = \lambda g_m - \kappa \log p$ are non-increasing in m by (B2), so φ_λ is concave, hence unimodal with maximum at m_λ by (60); a step towards m_λ therefore does not decrease φ_λ , that is does not decrease w_λ .

So every step of F_λ is w_λ -increasing. In particular no $A \neq P_{m_\lambda}$ is a strict local maximum of w_λ for flips and exchanges, since $F_\lambda(A)$ is one of its neighbours; and w_λ restricted to $2\gamma^*$ is proportional to π_λ there by (48). The two remaining claims of (70) are proved with Theorem 6.1. $\square$

PROOF OF PROPOSITION 6.6. Fix γ and $j \in S^c \setminus \gamma$ and write $\Pi = \Phi_{\gamma \cup \{j\}} - \Phi_\gamma$. By (6), $\psi_j(\gamma) = -\alpha \log p + \frac{n}{2} \log\{(g^{-1} + 1 - R_\gamma^2)/(g^{-1} + 1 - R_{\gamma \cup \{j\}}^2)\}$, and the computation (99)–(100), which used only that Π is a rank-one projection and Lemma A.1, gives

$$\psi_j(\gamma) \leq -\alpha \log p + \frac{4\|\Pi Y\|_2^2}{\sigma_0^2},$$

by (11). Now $\Pi Y = \Pi X_{\gamma^*} \beta_{\gamma^*}^* + \Pi \tilde{w}$ by (81), so by (82), Lemma A.5 applied to $\gamma \cup \{j\}$ and j , and Lemma A.2b applied to the nested pair $\gamma \subset \gamma \cup \{j\}$,

$$\|\Pi Y\|_2^2 \leq 2\|\Pi X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 + 2\|\Pi \tilde{w}\|_2^2 \leq 2n\omega(X)\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 + 4(L + \tilde{L})\sigma_0^2 \log p,$$

and $\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 \leq \|\beta_{\gamma^*}^*\|_2^2$. Multiplying by $4/\sigma_0^2$ gives (80). $\square$